OctoPus • OctaveMelody

OctoPus User Manual

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1. Introduction

OctoPus (Octave + Jianpu) is a desktop editor for numbered music notation. Write a score in JPS (Jianpu Script), review its live SVG preview, compare it with an original image or PDF, and export finished pages. A .jps file holds the music; page settings control its presentation. Find the installed version in the About dialog. This guide was updated 1 November 2026.

After installation, editing, rendering and supported transcription run locally. Help links open in the system's default browser, and update checks need a network connection. The manual is available offline in English and Chinese, with matching topics and examples. Use the language link on each topic page or download the PDF you need.

> Application screenshots are based on the Windows desktop UI. Menu labels and system fonts may differ on macOS and Linux.

Install and start OctoPus

Download a release package from the official OctoPus Releases page and choose the package for your operating system. Packages include the examples and offline documentation. After installation, scores, rendering and transcription work offline; update checks need a connection.

Platform	Install and launch
Windows	Run the setup executable and follow its prompts, or extract the portable ZIP and launch octopus.exe. Keep the portable folder together. Windows requires WebView2, which is already present on most systems.
macOS 26 or later (Apple Silicon)	Open the DMG and drag OctoPus to Applications, then launch it there. The current desktop package may be unsigned and not notarized, so Gatekeeper may block launch. See the Quick Start's macOS steps for the current release instructions.
Linux	For an AppImage, make the downloaded file executable and run it. Linux needs GTK/WebKit runtime libraries and Poppler tools.

Help → Quick Start includes a screenshot-led first score walkthrough after the app opens.

2. Capabilities, limits and roadmap

Available now	Scope
JPS editing	New/open/save, undo/redo, clipboard, find/replace, source formatting and bundled samples.
Live SVG preview	Source/score correspondence, page navigation, zoom and page layout controls.

Available now	Scope
Reference transcription	PNG/JPG/JPEG/PDF input, original display and provisional JPS drafts for review.
Export	SVG, PDF, PNG and JPEG; locally packaged worker and platform-specific score-font handling.

Current limits. Transcription may miss parts of a score. Check notes, accidentals, octave dots, durations, lyrics, slurs, grace notes, endings and page order. Filtering for repeated running headers, footers and page numbers is conservative, so unusual layouts and spans across pages need review. The desktop app currently has no audio playback or MIDI workflow, LilyPond export, MusicXML import/export, or color styling for generated sheet music. Staff notation is not currently supported as an input format; JPS remains the supported music source. Updates are not installed automatically. macOS desktop packages support Apple Silicon on macOS 26 or later; Windows/macOS host acceptance is separate from Linux testing.

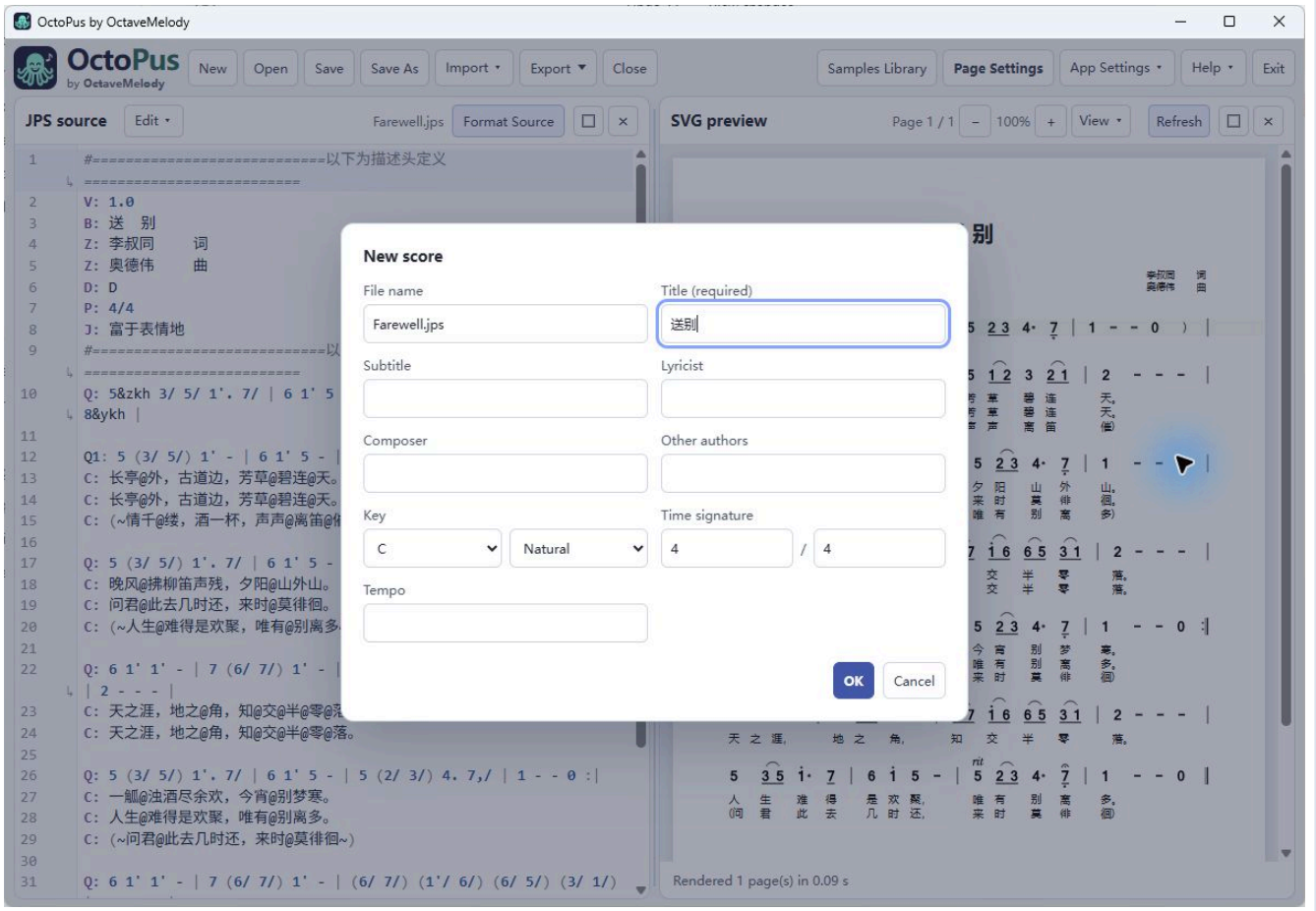
Planned features

- Add audio playback and MIDI import/export.
- Support MusicXML import and export.
- Export LilyPond source files.
- Add color support for generated sheet music.
- Continue improving image/PDF transcription, with optional LLM assistance.
- These plans are not current features; no release dates are set.

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3. Your first score

- Start OctoPus and choose **New**. Enter a title, choose the basic score fields and confirm. Time-signature values are positive integers; compound meters such as 12/8 are supported.
- Replace the source with the example below, or use **Open** to load a bundled example from the examples folder.
- Wait for the SVG preview to finish rendering. Click a note to locate its source; edit the text and review the result.
- Use **Page Settings** for margins/fonts, **Save As** for a .jps file, then **Export** for the finished score.



New score dialog: enter the title and basic score information before editing.

```
V: 1.0
B: First score
Z: Practice example
D: C
P: 4/4
J: 90

Q: 1 2 3 4 | 5 6 7 1' ||
C: 春 风 吹 来 花 儿 开 了
```

This example has two four-note bars; the final note is one octave above the base register. The eight Chinese lyric characters map to the eight notes.

Learn from a library score

Choose Open and select Farewell.jps from the examples folder, or download the excerpt below. Read the header and find the row beginning Q: 5 (3/ 5/). Place the source cursor on 1'. and inspect the dotted high note in the preview. Change only the dot, compare the result, then Undo. In a copied lyric row, move an @ and see which note loses its syllable. Use Save As before keeping your experiments. Downloadable excerpts include the original header, even when the page shows only body lines.

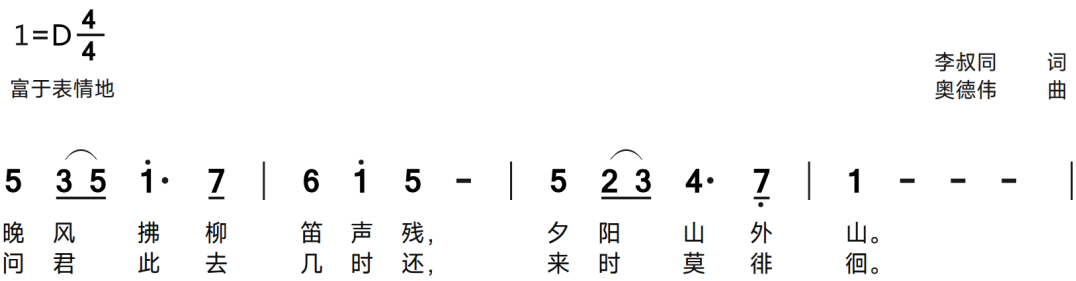
A melody with two lyric verses

Open Farewell from the examples folder, or download the excerpt below. This excerpt is its second sung row. Compare the four barlines, the dotted high 1, the lower 7, and the two lyric lines. Download the excerpt to try the same passage without the rest of the score.

bundled examples: Farewell.jps • source lines 17-19. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

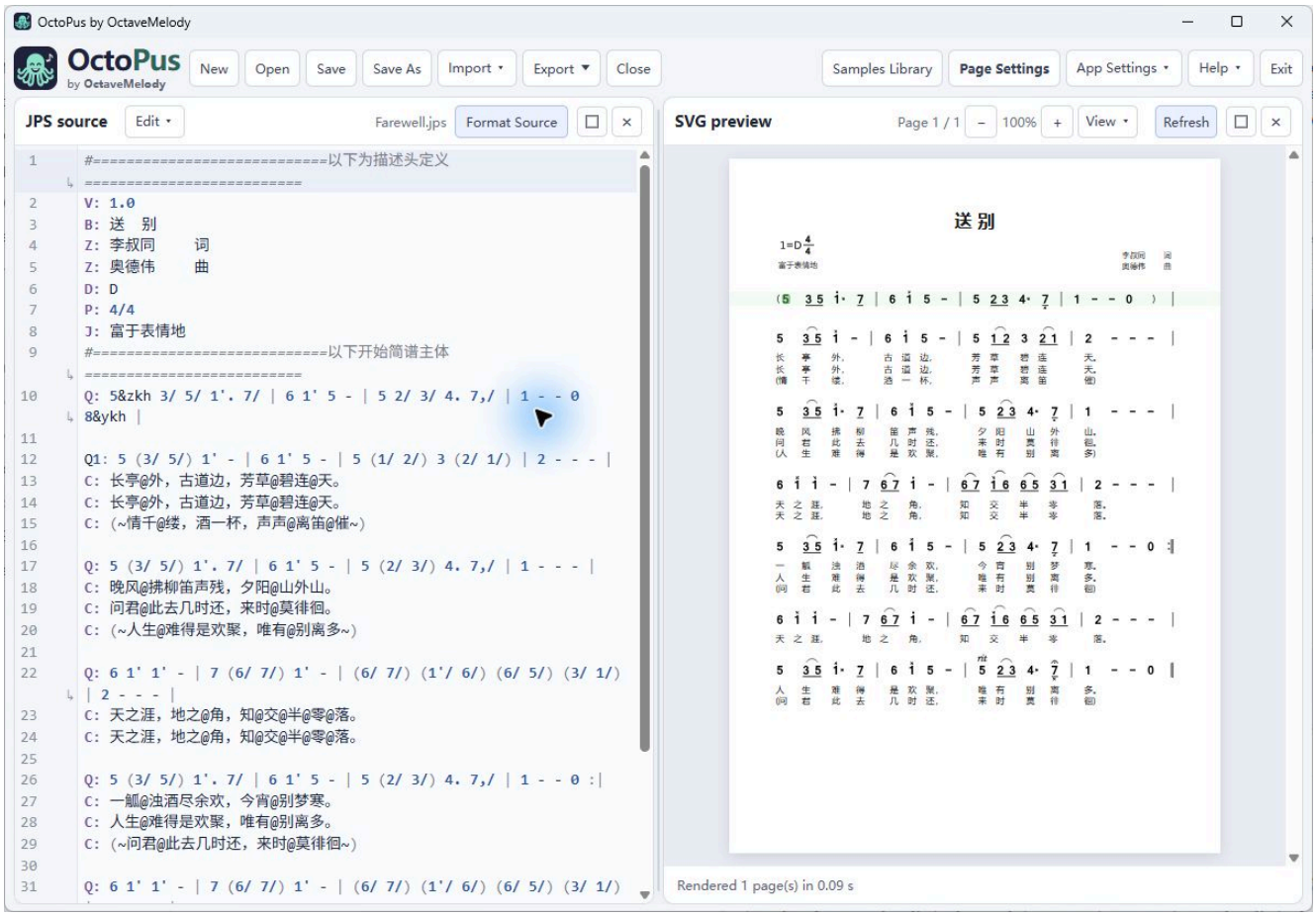
```
Q: 5 (3/ 5/) 1'. 7/ | 6 1' 5 - | 5 (2/ 3/) 4. 7,/ | 1 - - - |
C: 晚风@拂柳笛声残，夕阳@山外山。
C: 问君@此去几时还，来时@莫徘徊。
```



Rendered output: A melody with two lyric verses

4. The workspace and UI

The top row is ordered New, Open, Save, Save As, Import, Export and Close. Samples Library appears when enabled in App Settings; Page Settings appears while a score is open. App Settings, Help and Exit follow in that order. App Settings groups the N1/N2 Normal and T1/T2 Transcription layouts with sample-library visibility, interface language and UI-font choices. On first use, the interface defaults to Simplified Chinese and the platform font; OctoPus saves later choices. The menu marks each platform's default UI font: Microsoft YaHei on Windows, PingFang SC on macOS and bundled Noto Sans SC on Linux. The alternatives are SimSun, Songti SC and bundled Liberation Sans, respectively. UI fonts affect controls and menus; Page Settings controls score and export font roles.



Main workspace: edit JPS source on the left and review the matching score preview on the right.

Import offers a single JPG / PNG / PDF action. Transcription uses the bundled Default OCR profile. The logo and brand name open the project home. Each panel has Maximize/Restore and Close controls; drag a border between panels to resize them.

Pane	Purpose
JPS source	Edit text. The filename and Unsaved status appear just before the right-aligned Format Source button. Editing actions sit beside the JPS source heading.
SVG preview	Rendered score, fit/zoom/page controls and render diagnostics.
Original Image/PDF	Reference page, fit/zoom/rotate controls and Transcribe menu; visible in Transcription mode.

In a narrow pane, view controls move into a **View** menu and source tools into **Edit**. SVG View includes Fit Page, Fit Width, Prev Page and Next Page; Original View includes fit and rotation, plus PDF page navigation when applicable. Very narrow panes also provide zoom through View. These controls stay on the pane's heading row.

5. Files, saving and recovery

Open loads a .jps score. If you edit the score or page settings while another file is loading, OctoPus keeps your changes and asks you to retry. It supports plain JPS text and compatible JSON-wrapped JPS documents; a wrapper may contain source, page settings and existing metadata. **Save** writes the current document. A new document needs a destination first, while **Save As** changes the destination. To preserve page settings from a plain-text score, you may need to save it in a wrapper; follow the dialog prompt.

Close closes the score; **Exit** closes the application. If changes are unsaved, choose Save, Discard or Cancel. After an interruption, recovery can restore the last recoverable draft and managed references, but it does not replace regular backups. Reference images and PDFs are separate from the score source, so keep their original files. Opening a sample creates a working document; choose your own save location before overwriting anything.

Use **Open** to reopen a saved score or a bundled example. Examples are provided in the examples folder; downloadable excerpts in this manual can also be opened directly.

6. Editing JPS source

Use Undo/Redo, Cut/Copy/Paste, Find, Replace and Select All from the heading or compact Edit menu. Find opens the search UI; Replace exposes replacement controls. Standard editor keyboard shortcuts remain available. Clipboard availability depends on the operating system. **Format Source** spaces musical tokens while retaining notation content; it cannot repair musical mistakes or validate a transcription. Quoted annotations and attached grace groups need careful review after edits.

Manually selecting JPS text shows a blue selection, which becomes muted when the editor loses focus. Clicking a preview note marks its corresponding source note in green.

Prefer ASCII punctuation for syntax: colon :, quote ", comma , apostrophe ' and slash /. Use full-line # comments to explain the source. Do not paste curly quotes or a full-width colon as syntax. Keep attached modifiers next to their note. Preserve a copy before a large replacement and use Undo when needed.

Syntax checks appear while you type: gutter symbols and underlined source ranges mark errors or warnings. Select a gutter marker to jump to its range. A small ↳ marks a wrapped continuation of the same source line, not a new music row. Formatting applies only to valid music prefixes with a colon; descriptive text such as “Queen: some value” keeps its spaces.

7. Preview, cursor and navigation

The preview updates as source changes. Clicking or moving the source cursor highlights the matching score position and brings that row toward the preview center. Clicking a preview note or lyric selects its source without automatically moving the preview. Direct source clicks/keyboard movement resume following. Only uniquely mapped elements are selectable; some decorative or temporary-voice elements may not have a precise source target.

Use -/+ to zoom, Fit Page for the whole page and Fit Width for width-based reading. Page arrows or View menu items navigate rendered pages. Refresh requests a new render. Click diagnostics below the preview to jump to the related source. A previous preview can remain visible during rendering; wait for completion and resolve

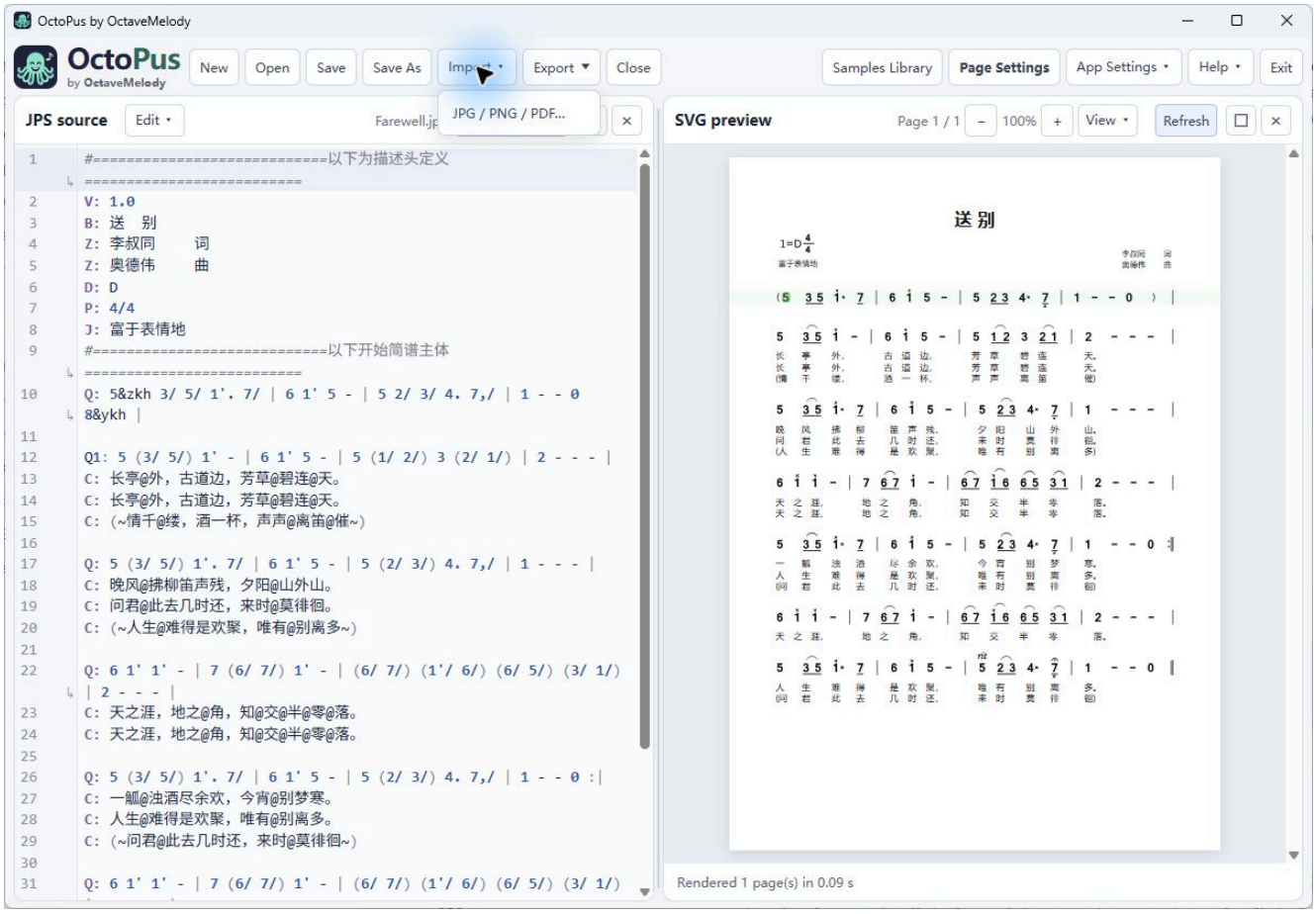
relevant warnings before exporting.

Focus the preview with Tab to use **Page Up** / **Page Down** for navigation and **Ctrl+wheel** for zoom. The status line reports completed pages and elapsed rendering time; during a multipage render it shows progress.

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8. Original images/PDFs and transcription

Choose **Import → JPG / PNG / PDF…** or drop **one** PNG, JPG/JPEG or PDF into Original Image/PDF. An existing reference prompts Replace Reference or Cancel. Import is blocked while another import, replacement decision, recovery, save, transcription or conflicting dialog is active. Unsupported/multiple dropped files are rejected without import.

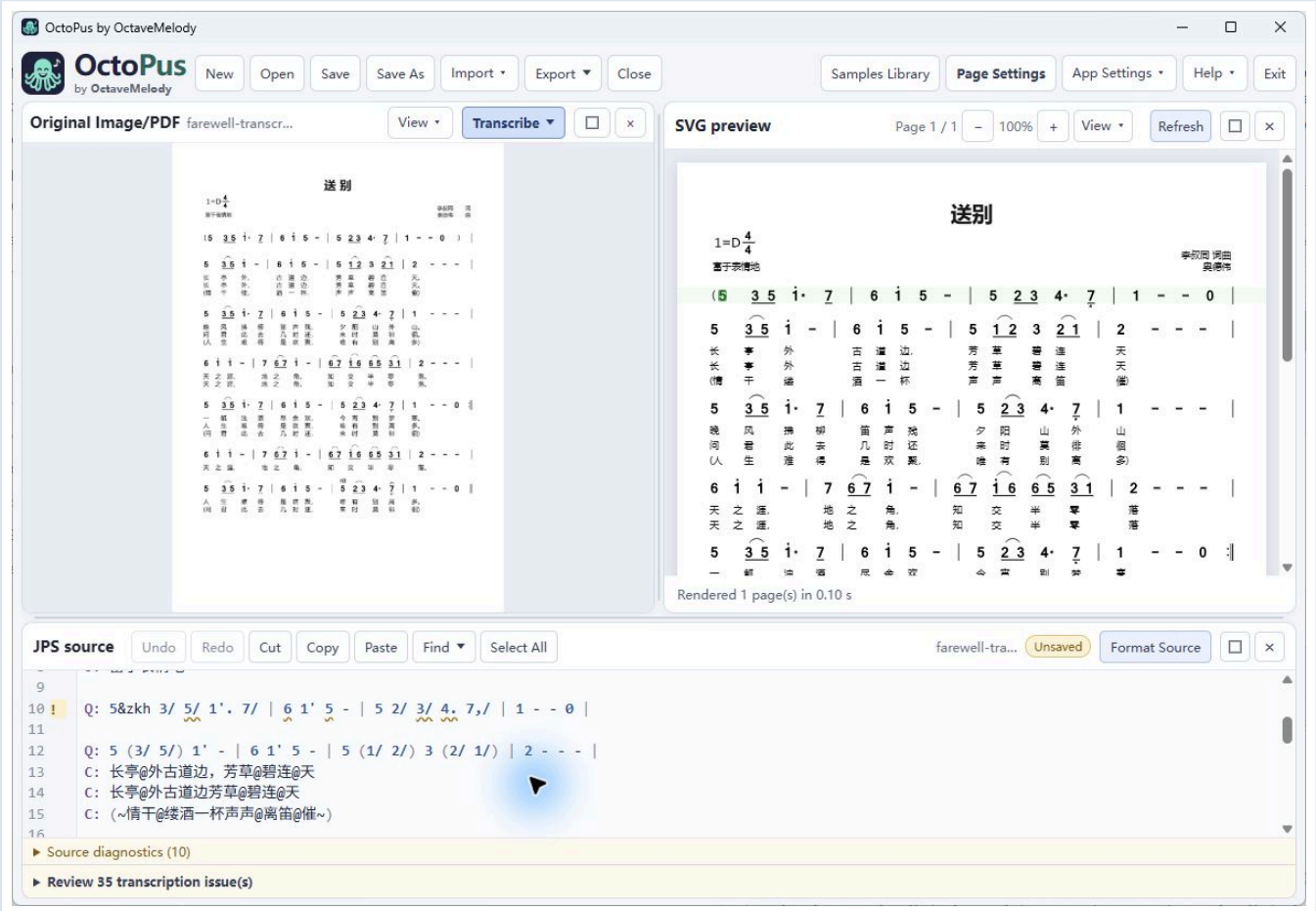


Import menu: choose JPG / PNG / PDF to add a single reference image or document.

Drag the displayed original to pan; use fit, zoom and Rotate 90°. Rotation changes the viewing orientation, not the original file. A PDF has page navigation. Closing the pane hides it and retains the managed reference; select a Transcription layout to reopen it. Input limits include 100 MiB per image/PDF, at most 200 PDF/reference pages and 40 million pixels per decoded image/PDF page. Damaged, encrypted or unsupported files may fail; reduce the input if the program reports a size or page-limit error.

- Transcribe New: generates a provisional new score. Handle any unsaved-score prompt before replacing the working score.

- Transcribe Append: adds recognized music to the current source, using a page break boundary when needed and omitting repeated draft headers for a nonempty score. Undo can reverse the insertion.
- A PDF is transcribed as a document, not only its displayed page. Review the warnings and each page against the original; collapse warnings when you need editing space.
- Correct uncertain recognition manually. Check octave dots, beams, dotted durations, accidentals, grace notes, slurs, endings and lyrics before saving/exporting.



Transcription review: keep the original visible while comparing the rendered draft and source.

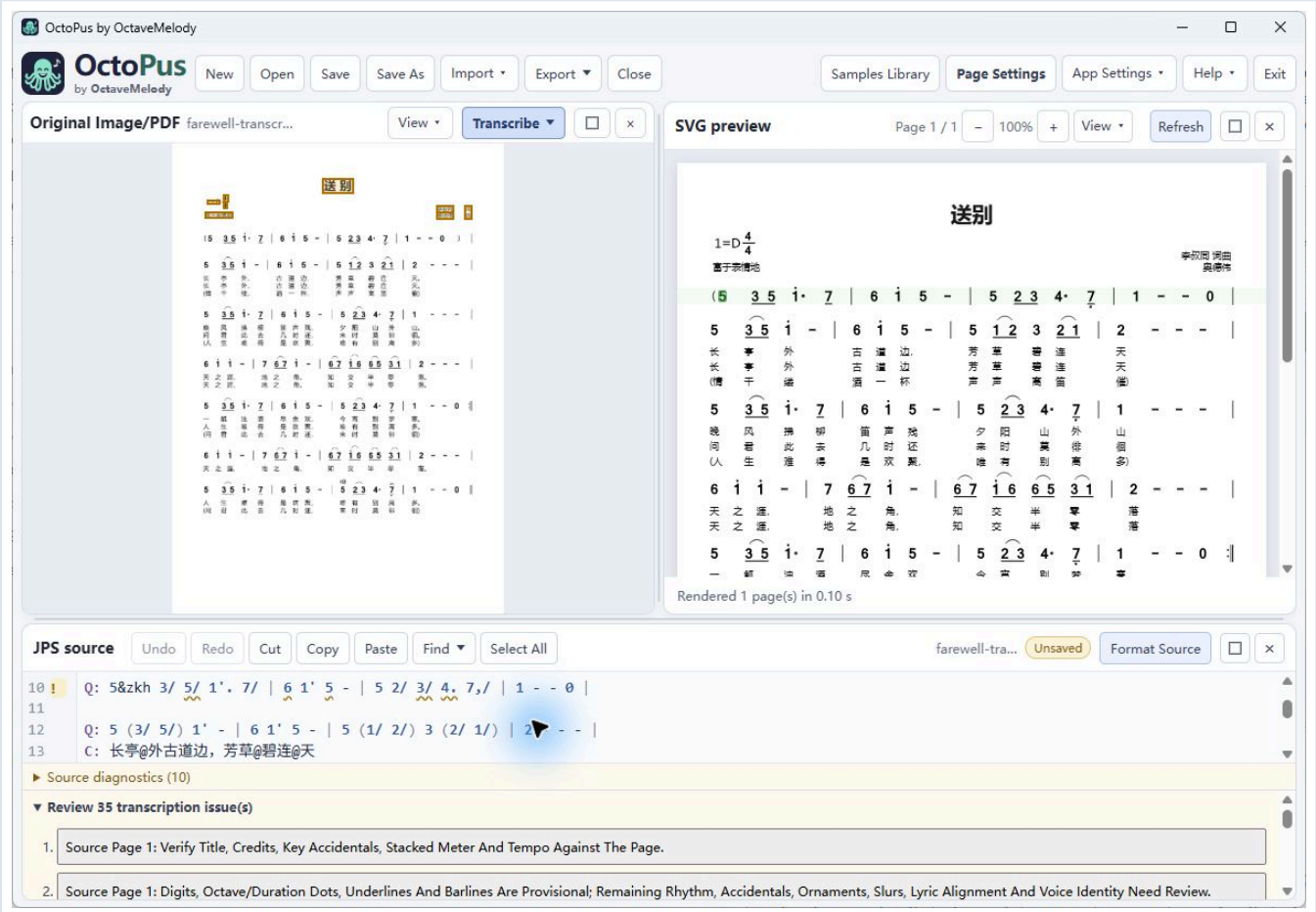
Before inserting into a nonempty score, Append compares each nonblank explicit uppercase D/P/J draft header with the current score. It blocks a definite mismatch or a header it cannot verify, such as a missing destination value or an ambiguous D/P/J instruction inside music, and suggests Transcribe New. It does not infer omitted draft headers or add default key, meter or tempo values. Matching headers do not prove musical compatibility; review the draft against the original.

OCR model

The bundled **Default** profile uses PP-OCRv6 Small text detection and recognition models with the shared PP-OCRv4 Mobile orientation classifier. It supports English and Chinese and works offline. Transcribe New and Transcribe Append use this same profile for every crop and PDF page. The profile does not change or become part of the saved JPS score.

Review an uncertain draft

For PDFs, transcription shows page progress and a **Cancel transcription** button. Cancel stops the task, keeps your current score and discards the unfinished draft. Afterward, expand the review list and choose a finding. The original pane opens the matching page, zooms toward the reported region and outlines it in amber. If a finding maps to one source note, that note is selected in the editor. Low digit agreement adds a warning mark to the draft; this score compares classifier results and is not a calibrated probability or a guarantee of correctness.



Review finding: select an item to open and highlight its location in the original.

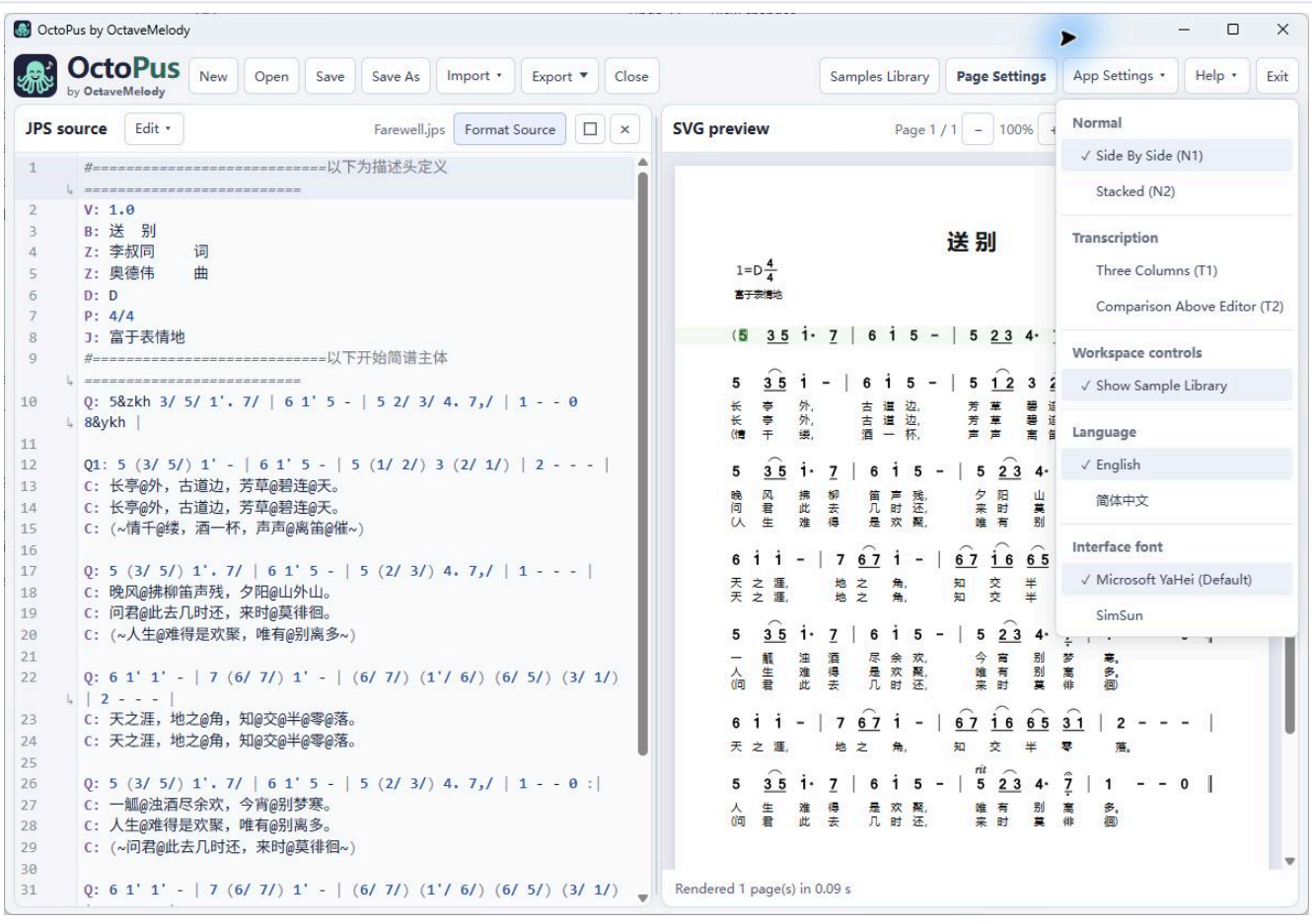
Editing the source clears obsolete note-range marks, but scan regions remain available for review. Old findings cannot select references that have been renamed, removed or replaced. A one-time hint explains how to drop in a scan or PDF and create a draft; dismiss it to clear the workspace.

OctoPus conservatively filters repeated headers, footers and isolated page numbers outside music and lyric bands. Review findings can flag continuation-page meter changes, key or tempo changes that need manual handling, unresolved score instructions, and clear measure or voice duration mismatches. Lyric findings point out differences between accepted words or slots and the rendered draft; review them manually because they neither rewrite lyrics nor verify OCR. Checks against the rendered draft compare note count, row and spacing, but none of these checks rewrites uncertain notes or proves that a draft is complete or musically correct. Compare the notation with the original before accepting it.

9. Modes, layouts and focus

Mode/layout	Use
Normal N1	JPS source and preview side by side.
Normal N2	JPS source and preview stacked.
Transcription T1	Original, source and preview in three columns.
Transcription T2	Original and preview above the source editor.

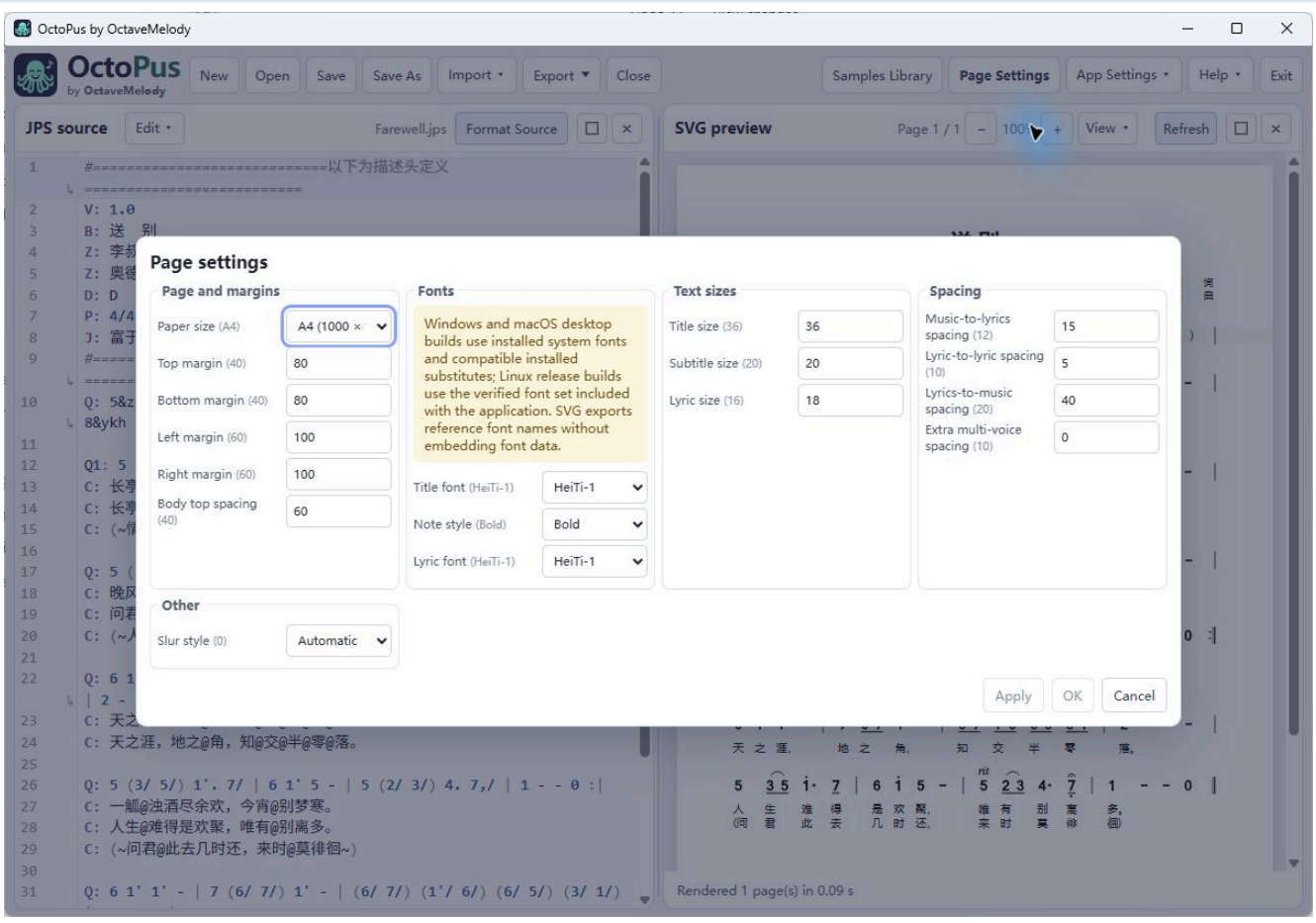
Drag the divider between panels to resize them. Vertical dividers move left/right; horizontal dividers move up/down. In T1, each divider resizes only its two neighboring panes. Each layout remembers its sizes. To resize with a keyboard, focus a divider with Tab and use the matching arrow keys (Shift for a larger step); Home/End select the smallest/largest size. Minimum sizes keep every panel accessible. Use a panel’ s Maximize control to expand it and Restore to return to the layout. Close hides a panel while preserving its content. Choosing a layout in App Settings reopens its panels; if all panels are closed, use that menu to resume. Controls adapt as the available width changes. F11 toggles the window maximized state; the toolbar has no separate Maximize button.



App Settings: switch between side-by-side, stacked and transcription layouts.

10. Page settings and fonts

Page Settings groups paper (A4/A5), margins and body offset; title/lyric fonts and note style; title/subtitle/lyric sizes; music-to-lyric, lyric-to-lyric, lyric-to-music and voice spacing; and slur shape (Automatic, Arc, Flat top). Sizes/margins are renderer units, not millimeters. **Apply** updates the score while keeping the dialog open; **OK** applies and closes; **Cancel** discards unapplied draft changes. Already applied changes are not reverted by Cancel. Save the score to retain changes.



Page Settings: adjust score page size, spacing, text size and notation fonts.

Role	Windows installed family	macOS installed family	Linux bundled family
HeiTi-1 (default)	Microsoft YaHei	PingFang SC	MiSans Regular
HeiTi-2	SimHei	Heiti SC	LXGW Neo XiHei
SongTi	SimSun	Songti SC	SimZhiSong
KaiTi	KaiTi	Kaiti SC	LXGW WenKai Regular
FangSong	FangSong	STFangsong	Zhuque Fangsong Regular

App Settings selects the interface language and UI font; see section 4 for the first-use defaults. These fonts affect controls and menus only. **Page Settings** selects score font roles and styles, while exports use the same roles. On Windows and macOS, OctoPus uses the preferred installed family for each role. If it is missing, OctoPus

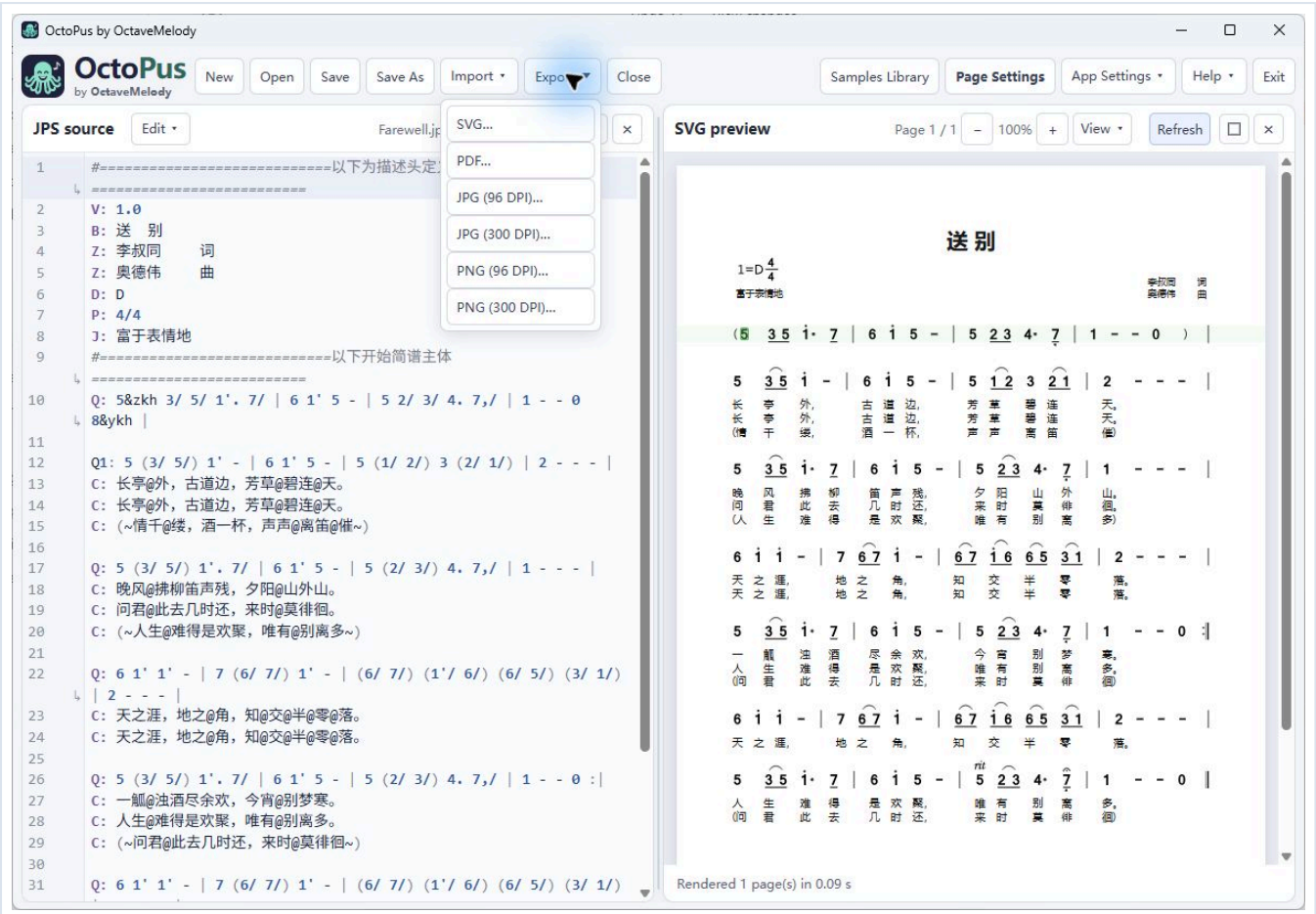
can use a compatible installed substitute. Open **Help → Font status** for official installation instructions or choose **Check again**. Windows links to Microsoft's supplemental-font guidance; macOS links to Apple Font Book. You can keep using installed substitutes without installing more fonts, though a character may be unavailable if none of the installed fonts contains it. Windows and macOS desktop packages do not include the 13-face score-font set. The Linux AppImage uses the verified font set shipped with the application. Noto fonts provide Linux glyph fallbacks, but they do not cover every Unicode character; Latin Arial-family text uses Liberation Sans. Note styles are Regular, Italic and Bold, and saved HeiTi maps to HeiTi-2. Font substitution can change spacing. Microsoft fonts are not redistributed. MiSans has a separate Xiaomi license; other Linux score-font notices use OFL and IPA licenses. The offline manual includes only small Noto Sans SC subsets. Microsoft font installation instructions • Apple Font Book instructions.

Image and scanned-PDF transcription uses the bundled Default OCR profile with RapidOCR + ONNX locally. Select **Transcribe** in the original-image/PDF panel to create a reviewable JPS draft. Review the generated source and any issue markers before saving or using the draft.

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11. Export and Windows portable use

Choose Export, select SVG, PDF, PNG or JPG (96 or 300 DPI), and choose where to save the files. PDF creates one multipage document; SVG, PNG and JPG create separate page files. Use 300 DPI for print-quality raster output. Save the JPS as your editable master before sharing exports. PDF embeds fonts, PNG preserves lossless pixels, JPEG compresses pixels and SVG keeps vector content. External SVG viewers still need matching text fonts, so inspect exports in the viewer you plan to use. The separate Python ReportLab PDF path cannot embed CFF-outline system fonts and uses the role fallback; this limitation does not apply to native desktop export.



Export menu: select a format and resolution, then choose where to save the result.

For the Windows portable ZIP, keep the extracted folder together, including engine, examples and docs, and run `octopus.exe` from that folder. Its worker is bundled, so no separate Python is required. Windows still requires WebView2.

Batch export from the Python command line

For a folder of JPS sources, the Python command line can produce one combined PDF or separate files per score. Files are processed in sorted path order; recursive output keeps subfolders. Existing output files are never overwritten. A failed source is reported while other valid sources continue; review the exit status and printed summary.

```
uv run --locked octopus batch-export /path/to/scores --out combined.pdf
uv run --locked octopus batch-export /path/to/scores --out-dir exported --format pdf --recursive
uv run --locked octopus batch-export /path/to/scores --out-dir images --format png --dpi 300
```

These commands use the Python installation described in ENVIRONMENT.md. They are not arguments to the portable desktop launcher.

12. Keyboard and Help

Shortcut	Action
Ctrl/Cmd+N	New score
Ctrl/Cmd+O	Open score
Ctrl/Cmd+S / Shift+S	Save / Save As
Ctrl/Cmd+Z	Undo
Ctrl/Cmd+Y or Shift+Z	Redo
Ctrl/Cmd+X / C / V / A	Cut / Copy / Paste / Select All in the editor
Ctrl/Cmd+F	Editor search
Ctrl+H (Windows/Linux), Cmd+Option+F (macOS)	Replace in editor
Page Up / Page Down; Ctrl+wheel	Navigate / zoom the focused preview
F11	Toggle window maximization
Arrow / Home / End / Escape	Navigate menus; Escape dismisses.

Some shortcuts are inactive while a modal dialog is open or a form control has focus. Help includes Quick Start, User Manual, Report Issues, Submit Requests, Check for Update and About OctoPus. Quick Start and User Manual open in the current interface language; Download PDF opens the native save dialog in the desktop app and downloads the file in a browser. Report Issues opens the project issue page, and Submit Requests opens its pull-request page. Check for Update compares the latest stable GitHub release and reports unavailable information, unfamiliar tags and network errors separately. It does not install updates. About and the logo link to the project home.

About. The dialog shows the manually assigned SemVer version, the number of commits in OctoPus product history and the short SHA of the product commit used for the build. It also displays “OctoPus: Oct(Octave) + Pus (JianPu) = The ultimate numbered notation editor.” Acknowledgments name Tomato JianPu for the adopted JPS format and rendering-compatibility goal, RapidOCR (Apache-2.0), ONNX Runtime (MIT), and the authors of the Linux score-font set and offline-manual Noto Sans SC subsets. The score-font set includes MiSans, LXGW Neo XiHei, SimZhiSong, LXGW WenKai, Zhuque Fangsong, Noto Sans SC, Noto Serif SC and Liberation Sans. Windows and macOS desktop builds use system fonts instead of these bundled score fonts. OctoPus is released under the GNU General Public License, version 3. The software is free of charge; the author provides no warranty and accepts no liability for direct or indirect losses arising from its use. Third-party licenses for included assets are bundled. Planned features are listed in section 2.

13. JPS structure and conventions

JPS is ordinary text, normally stored with extension .jps. A score has a descriptive header followed by music Q: and lyrics C:. One Q line represents a printed music row; following C lines belong to that row. Numbered prefixes identify voices. A blank line ends the current system group and starts another; it is a structural separator, not

just extra readability whitespace. A whole line beginning with # is a comment; inside a Q line, # is an accidental, not a comment.

The original specification calls V, B, D and P required fields. Include them for a complete portable score even though OctoPus tolerates incomplete drafts. Use uppercase header/prefix letters and the exact, case-sensitive lowercase literal `[fenye]`. Preserve the header/body distinction; repeated B/Z fields serve specific layout purposes. Unknown syntax may be preserved with diagnostics rather than rendered as intended.

Read a score in three passes

First read the header for title, key, meter and tempo. Then read only Q rows: digits and rests are musical events, suffixes modify their host, and barlines divide the flow. Finally read C rows and count lyric slots against the notes. The source order is also layout order: moving an entire Q row changes where music appears; moving its C row away may attach the words to a different row. Start the body with music, not unattached lyrics.

Plain JPS versus a saved document wrapper

A library .jps can be a JSON document whose code field contains the JPS and whose page_config field stores layout settings. Edit the code in JPS source, not the JSON punctuation. Downloadable manual excerpts are plain UTF-8 JPS for copying and opening. They carry musical headers and body text; to reproduce the pictured spacing exactly, open the named library sample so its page settings are restored.

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14. Descriptive header

Field	Example value	Meaning
V	1.0	Script version / 脚本版本
B	Practice	Title; repeats become subtitles / 标题，重复成为副标题
Z	Name - composer	Author/credit; multiple right-aligned lines / 作者，可多行右对齐
D	E$	Key; A-G with # sharp or $ flat / 调号，A-G 加 # 或 $
P	4/4	Meter; alternatives may follow / 拍号，可附其他拍号
J	90	Tempo number and/or descriptive text / 速度数值及/或文字
X	10	Score sequence number / 乐谱序号

```
V: 1.0
B: Main title
B: Subtitle
Z: Writer - lyrics
Z: Composer - music
D: F#
P: 4/4 (2/4 1/4)
J: 90 Allegro
X: 10
```

Use a capital A-G key letter: C, F#, E$. Multiple B lines put subtitle text under the first title; multiple Z lines place credits at the upper right. P uses slash-separated numerator/denominator; spaces and parentheses can describe auxiliary meters. J may be numeric beats per minute, words, or both. The spec defines X as a score sequence number, not a voice number; current OctoPus preserves X as header text but does not render it. Header values describe the score; a tempo field does not enable playback.

Required fields and repeated fields

For a complete score include V, B, D and P. V: 1.0 declares the syntax version; it is not the OctoPus application version. The first B is the centered main title and later B lines are centered subtitles. Each Z creates a separate right-aligned credit line. Optional J describes speed, and X can carry a catalog number. Keep these lines above the first Q. Empty or incomplete drafts may render, but do not rely on that as a portable document format.

Key, meter, tempo and comments

D: F# and D: E$ mean sharp and flat keys; use capital A-G and ASCII #/$, not full-width punctuation. P: 6/8 describes the meter; P: 4/4 (2/4 1/4) also lists auxiliary meters. It does not replace local barline meter changes. J: 90 uses a numeric tempo, J: Allegro uses text, and both can coexist. A full line starting # is ignored as a source comment, whereas 4# inside Q is a sharp note. Quoted annotations are printed; source comments are not.

Header placement and a multilingual title

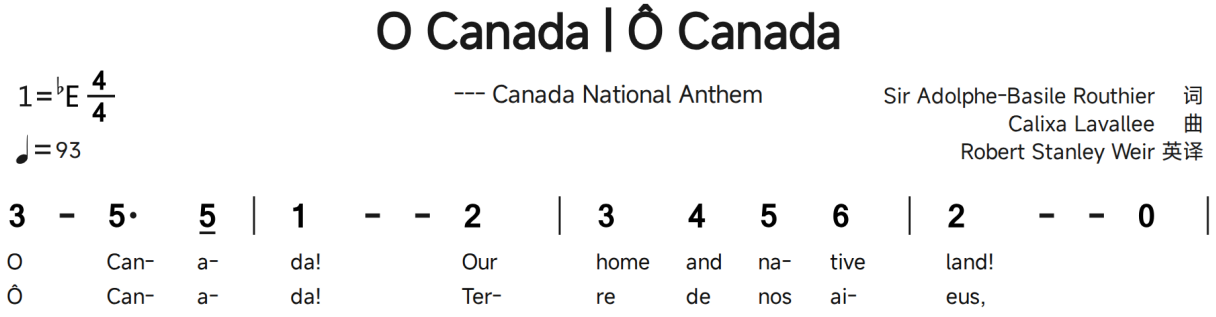
The second B is a subtitle; the three Z lines are separate credits. D: E$ displays the flat key. P sets 4/4 and J supplies a numeric tempo. The title punctuation is text, not a musical barline.

bundled examples: O-Canada.jps • source lines 13-15. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
V: 1.0
B: O Canada | Ô Canada
B: --- Canada National Anthem
D: E$
P: 4/4
J: 93
Z: Sir Adolphe-Basile Routhier　词
Z: Calixa Lavallee　曲
Z: Robert Stanley Weir 英译

Q: 3 - 5. 5/ | 1 - - 2 | 3 4 5 6 | 2 - - 0 |
C: O /Can- /a- /da~! Our/ home / and /na-tive/ land~!
C: Ô Can- /a- /da~! Ter- /re /de /nos /ai- /eus~,
```



Rendered output: Header placement and a multilingual title

15. Notes, register and duration

Syntax	Meaning
1-7	Scale degrees / 简谱音级
0	Rest / 休止符
8	Invisible rest with layout space / 占位隐藏休止符
9	Rhythm X note / 节奏 X 音符
1' / 1''	One/two octaves above / 高一/两个八度
1, / 1,,	One/two octaves below / 低一/两个八度
1 - -	Quarter note plus two extension beats / 音符加两拍增时线
1/ / 1// / 1///	Eighth / sixteenth / thirty-second / 八分、十六分、三十二分
1. / 1..	Single/double dot / 单附点、双附点

A bare digit is the basic quarter-note unit in the compatibility notation. Each attached slash halves its duration. A dot adds half the undotted duration; two dots add half plus a quarter (1.75 times the original). Extension hyphens add the Jianpu continuation strokes. They do not each need a lyric skip. Add octave commas/apostrophes after the digit; avoid spaces between the note and its suffix.

> Q: 1 2' 3,, 0 | 1/ 2/ 3// 4// 5/ 6 7 | 6. 7/ 1' - ||

Scale degree, rest, hidden rest and rhythm X

1-7 name scale degrees relative to D; they are not absolute MIDI pitches. 0 is a printed rest. 8 is an invisible rest/blank slot used to reserve layout space, especially between voices; it is not the next scale degree. 9 draws a rhythm X rather than pitch 9. Continuation - strokes extend a sounding note; they are not additional sung attacks. A hidden 8 is still a source event: when lyrics drift, compare its placement and any @ markers rather than simply deleting every invisible slot.

Calculate duration before editing layout

The basic digit is a quarter-note unit. Relative values are: no slash = 1, / = 1/2, // = 1/4, /// = 1/8. A single dot multiplies the value by 1.5 and a double dot by 1.75. Thus 5/. is 3/4 of a quarter-note unit and 5//. is 3/8; 5.. is 1.75. Each separate continuation dash adds a continuation beat in ordinary Jianpu writing. In 4/4, four bare notes fill an ordinary bar; in 6/8, six eighth notes do. Beam joins, lyric joins and extra spaces do not fix an incorrect duration total.

Keep suffixes attached to their host

Use apostrophes for high register and commas for low register; repeat them for two or more octaves. A note such as 7,/# carries low register, an eighth-note slash and a sharp. Library scores use several compatible suffix orders, so start from a working example instead of splitting a compound token into disconnected text. A period inside music is a duration dot; punctuation inside C is lyric punctuation. Do not replace the ASCII apostrophe with a typographic quote.

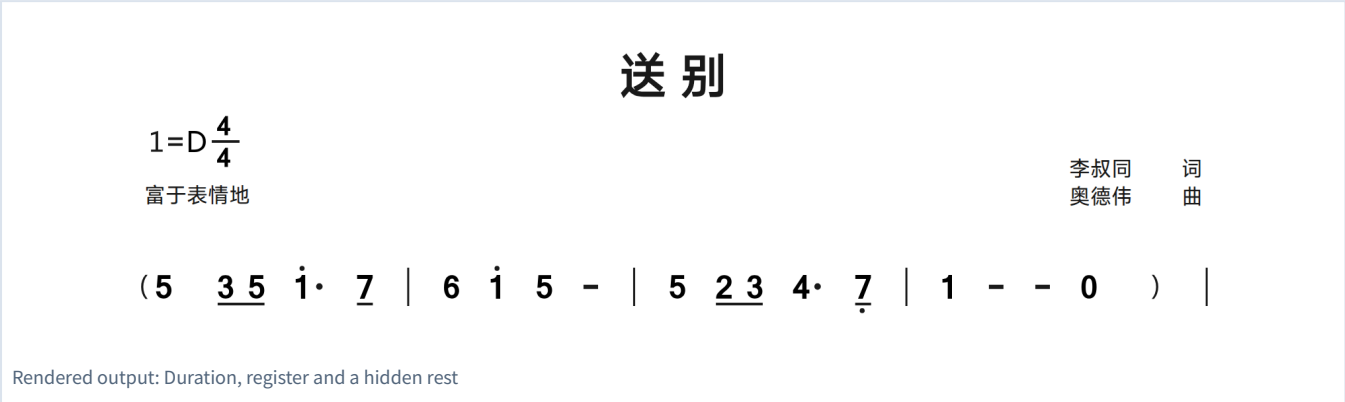
Duration, register and a hidden rest

Read 3/ and 5/ as eighth notes, 1'. as a dotted high note, and 7,/ as a low eighth note. The final 0 is visible; 8 reserves space but prints no rest digit. &zkh / &ykh delimit the accompaniment visually.

bundled examples: Farewell.jps • source lines 10-10. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 5&zkh 3/ 5/ 1'. 7/ | 6 1' 5 - | 5 2/ 3/ 4. 7,/ | 1 - - 0 8&ykh |
```



Rendered output: Duration, register and a hidden rest

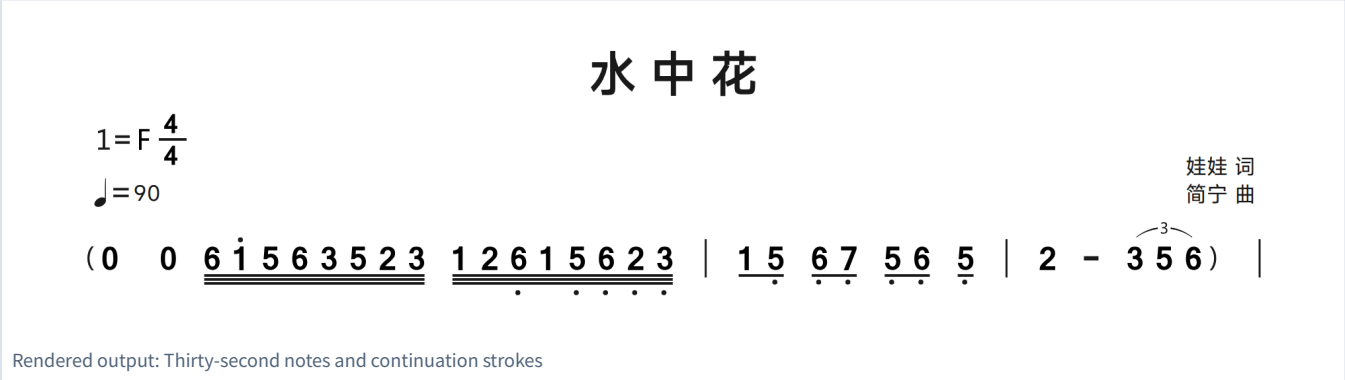
Thirty-second notes and continuation strokes

Three slashes produce three shortening lines. Compare the opening 0 rests, the low commas and high apostrophes, the short-note run, and 2 - at the end. The final (y...) is a tuplet, explained later.

bundled examples: Flower-In-Water.jps • source lines 10-10. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 0&zkh 0 6/// 1'/// 5/// 6/// 3/// 5/// 2/// 3/// 1/// 2/// 6,/// 1/// 5,/// 6,/// 2,/// 3,/// | 1/ 5,/ 6,/ 7,/ 5,/ 6,/ 5,/ | 2 - (y3 5 6&ykh) |
```



Rendered output: Thirty-second notes and continuation strokes

16. Automatic and custom grouping

OctoPus automatically groups short notes by a dotted-quarter beat in compound eighth-note meters: 6/8 uses two groups of three eighth notes, 9/8 uses three, and 12/8 uses four. The same rule applies when the denominator is 8 and the numerator is greater than 3 and divisible by 3. Other meters, including 3/8 and 2/2, retain quarter-note grouping. The note durations themselves do not change.

```
P: 6/8
Q: 1/ 2/ 3/ 4/ 5/ 6/ |
```

Where ~ and ^ act

The Jianpu-Spec describes `~` and `^` as temporary controls for grouping and splitting beats; current OctoPus uses them for that same purpose. The example above forms two three-eighth groups without extra marks. Use the symbols only to override automatic grouping: attach ~ to join the marked note to the following note, or ^ to end the group after the marked note. For example, `1/~2/^3/~4/^5/~6/` explicitly groups six eighth notes in pairs. These controls change the visual underlines, not pitches or durations. Inspect both underline levels in a sixteenth-note run. In lyrics, ~ instead joins text into one lyric slot. Neither music ~ nor ^ adds a note.

Join and split beams in compound meter

The header lists 6/8 and 9/8; the first meter sets the initial dotted-quarter grouping. Explicit marks can customize it: in the first row, 6/^ ends the group after that note, while 5/~ 4/~ 1/ joins a group. The second row uses ~ to join longer runs. These marks change beam grouping, not note lengths.

bundled examples: Autumn-Cicada - Fei-Yuqing.jps • source lines 11-12. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 6/&sby&zkh | 5 6/^ 5/~ 4/~ 1/ | (3'. 3') 3'/ | 2 3/ 2 1/ | (7,. 7,.) |
Q: 5,//~ 6,//~ 7,//~ 1//~ 2//~ 3// 1//~ 2//~ 3//~ 4//~ 5//~ 6// | 4//~ 5//~ 6//~ 7//~ 1'//~ 2'// 7//~ 6//~ 5//~ 6//~ 4//~ 2// | (1. 1.) 8&ykh |
```



Rendered output: Join and split beams in compound meter

---

17. Accidentals and grace notes

Write sharp #, flat $ or natural = after the host digit, for example 4#, 7$, 3=. Grace notes are attached square-bracket groups after their host: `1[23]` displays preceding grace notes; `1[h23]` displays following grace notes. Inside a group, commas/apostrophes, slashes and accidentals still apply. Grace digits default to eighth notes; a slash makes a sixteenth. A bracket attached to a note is not an ending bracket on a barline.

```
Q: 4# 7$ 3= 1[2/3'] | 5[h6,7/] - ||
```

Accidental placement and scope

Write the accidental on its host digit, such as 4#, 7$ or 3=. Its printed position is before the digit. D changes the header key; a note accidental changes a particular written note. When copying a passage into a different key context, inspect its later repeated notes and explicit naturals. OctoPus prints the notation supplied; it does not validate the intended harmonic spelling or infer every missing cancellation.

Grace notes are smaller attached groups

1[23] means a preceding group attached to main note 1; 1[h23] means a following group. The h goes just inside [, not after the closing bracket. The group may contain several notes, register marks, accidentals and shortening slashes. Inside a grace group a bare digit already has the short eighth-note form; / makes it sixteenth. The host still has its own duration and lyric. Do not count each grace digit as a separate main lyric slot, and do not move the brackets to a barline, where they become ending syntax.

Flat/sharp followed by a natural

In Q2, 3$ is followed by 3=; in Q4, 4# is followed by 4=. The modifier is typed after the digit but drawn before it. Hidden 8 slots keep the choir parts aligned. These two selected voices are rendered independently of the omitted parts.

bundled examples: Edelweiss - Choir ChinEng.jps • source lines 23-23, 27-27. Excerpt rendered with the Linux package font set; blank page space omitted.
Download this JPS excerpt

JPS source

```
Q2: 3 - 8 3 | 5 - - | 5 - 8 5 | 4 - - | 3 - 5 | 4 8 3$ 8 2 | 3= - - | 1' - - |
Q4: 1 - 8 1 | 7, - - | 3 - 8 3 | 4 - - | 3 - 5 | 5 8 4# 8 4= | 3 - - | 1' - - |
```



Rendered output: Flat/sharp followed by a natural

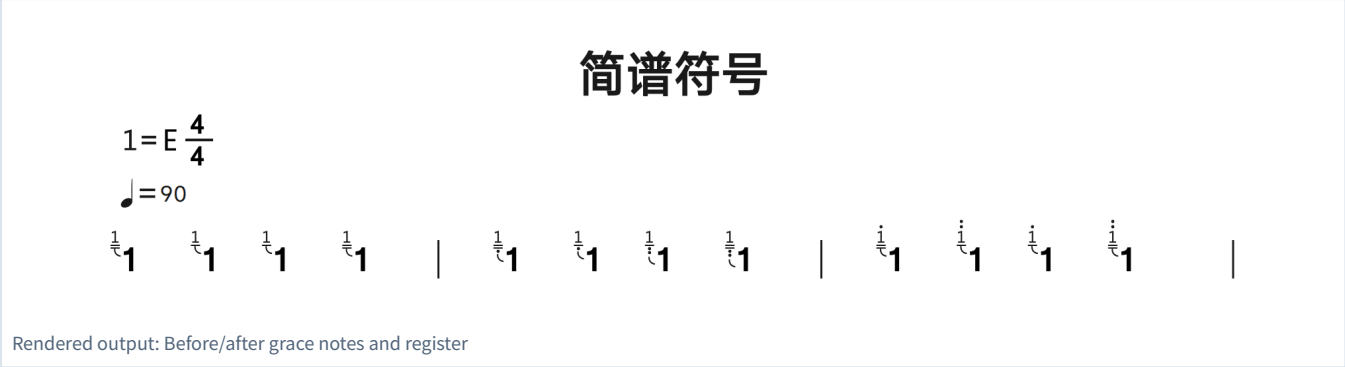
Before/after grace notes and register

The first four hosts compare before/after grace notes and their shorter values. The next bars compare low and high grace notes. [h...] follows the host visually; [...] precedes it. All groups are typed after the main note.

bundled examples: Symbols.jps • source lines 8-8. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 1[1/] 1[1] 1[h1] 1[h1/] | 1[1,/] 1[1,] 1[h1,,] 1[h1,,/] | 1[1'/] 1[1''] 1[h1'] 1[h1''/] |
```



Rendered output: Before/after grace notes and register

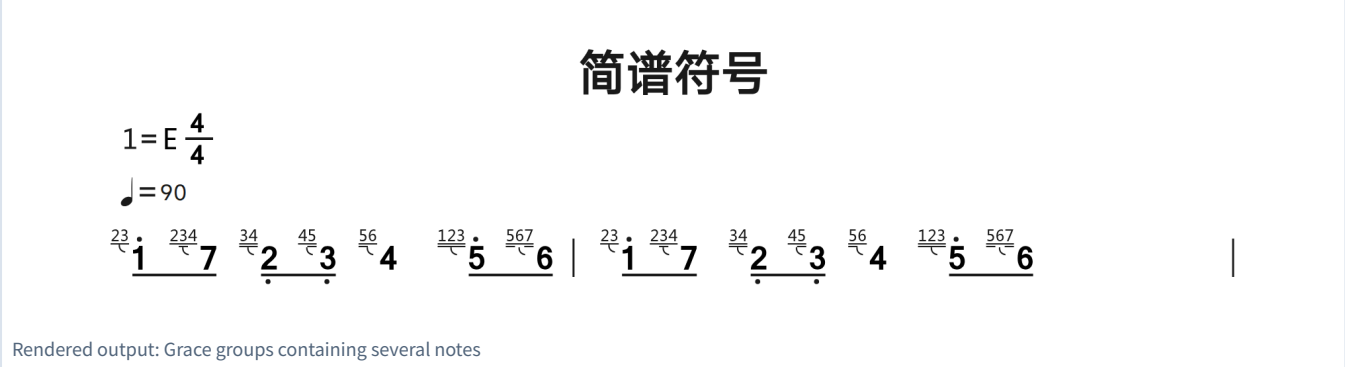
Grace groups containing several notes

The left bar contains preceding grace groups; the right bar repeats the pattern as following grace groups with h. Slashes within a group shorten the grace notes independently of the host.

bundled examples: Symbols.jps • source lines 15-15. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 1'/[23] 7/[23/4] 2,/[3/4/] 3,/[45/] 4[5/6] 5'/[1/2/3/] 6/[5/67/] | 1'/[h23] 7/[h23/4] 2,/[h3/4/] 3,/[h45/] 4[h5/6] 5'/[h1/2/3/]
6/[h5/67/] |
```



Rendered output: Grace groups containing several notes

---

18. Dynamics and hairpins

Attach an ampersand code to a note, such as `3&mp`. The spec uses `&tempo` for a tempo; OctoPus also accepts the legacy spelling `&a tempo`. The samples below show what this build actually draws, including accepted terms with no visible mark. Hairpins start with < (crescendo) or > (decrescendo) and end with !. Add + after the opening marker to move it upward when it overlaps another span. These are printed instructions; the desktop app does not interpret them as audio playback controls.

```
Q: 1&p< 2 3 4! | 5>+ 4 3 2! | 1&a tempo - - - ||
```

Code	OctoPus sample	Meaning	Current OctoPus rendering
&ppp		Extremely soft	Visible in this OctoPus preview
&pp		Very soft	Visible in this OctoPus preview
&p		Soft	Visible in this OctoPus preview
&mp		Moderately soft	Visible in this OctoPus preview
&mf		Moderately loud	Visible in this OctoPus preview
&f		Loud	Visible in this OctoPus preview
&ff		Very loud	Visible in this OctoPus preview
&fff		Extremely loud	Visible in this OctoPus preview
&cresc		Crescendo	No additional mark in this OctoPus preview
&dim		Diminuendo	Visible in this OctoPus preview
&sf		Sudden accent	No additional mark in this OctoPus preview
&fp		Loud then immediately soft	No additional mark in this OctoPus preview
&sfp		Accented then immediately soft	No additional mark in this OctoPus preview
&tempo		A tempo; return to the original speed	No additional mark in this OctoPus preview
&a tempo		Legacy spelling for a tempo	No additional mark in this OctoPus preview
&rit		Gradually slower (rit.)	Visible in this OctoPus preview

Start, end and height of a hairpin

Place < or > on the starting host and ! on the ending host. A lone opening marker has no dependable endpoint; close it deliberately before the page boundary. <+ or >++ raises that span above the default height, with more + requesting a greater adjustment. Use the smallest offset that avoids a slur/ending collision, then inspect nearby systems and exports. Page spacing controls move whole rows; a + offset moves the selected mark. These are different corrections.

Each crop shows the notation as rendered by this OctoPus build.

Mark	OctoPus sample	Role	Current OctoPus rendering
<		Crescendo starts here	Draws an opening wedge toward the next !
>		Decrescendo starts here	Draws a closing wedge at the next !
!		Ends an open hairpin	Closes the open span; no separate printed glyph
+		Raises the opening hairpin; repeat + for more height	Raises the wedge; + itself is not printed

Dynamics are printed instructions

`&ppp` and `&fff` use packaged vector outlines, like `&pp` and `&ff`. They share the same note attachment and octave/slur clearance without changing the note's duration or spacing. For example: `Q: 1&ppp - - - | 1&fff - - - |`.

Attach a term to the note where the instruction begins. Keep the legacy multiword form &a tempo intact when using it. Visible terms describe loudness, change or speed as listed; rows with no additional mark are recognized but not drawn by this OctoPus build. An &yc fermata is an ornament, not a loudness instruction, although it uses the same ampersand mechanism. Printed tempo/dynamics do not provide playback or MIDI output.

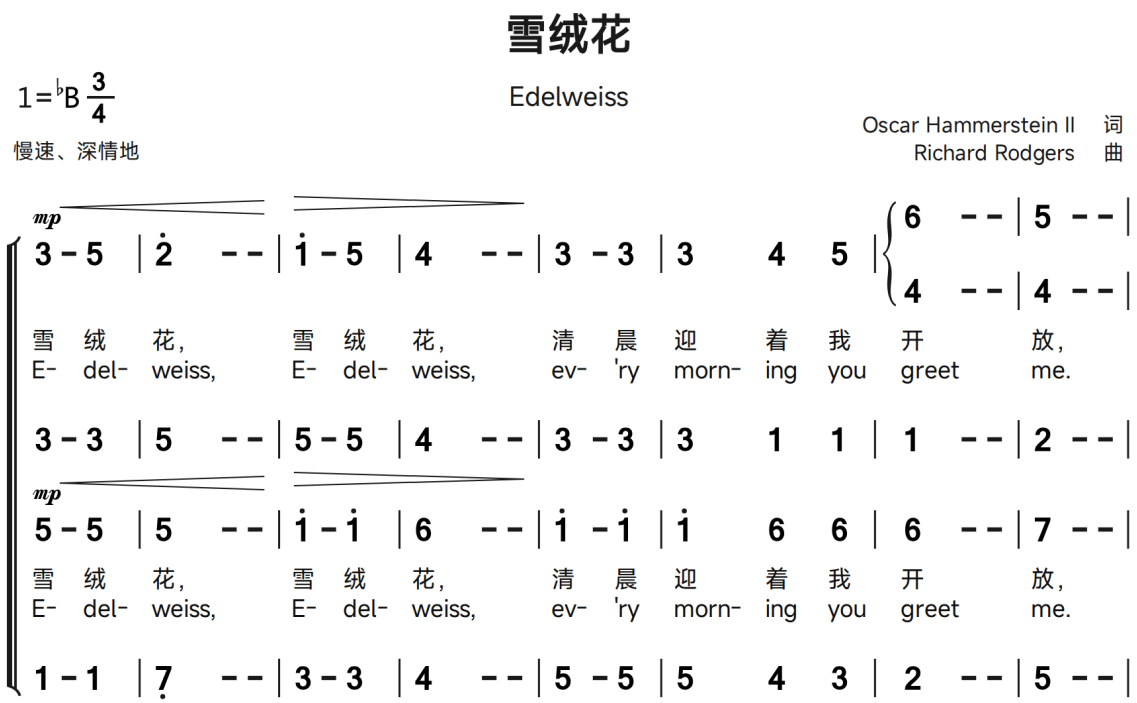
Dynamics and raised hairpins in a choir

Q1 and Q3 start with &mp. <+ begins a raised crescendo, ! ends it, then > begins a decrescendo. Compare the span endpoints with the notes and rests in the other parts; lyric rows remain attached to their source system.

bundled examples: Edelweiss - Choir ChinEng.jps • source lines 11-18. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q1: 3&mp -<+ 5 | 2' - -! | 1'> - 5 | 4 - -! | 3 - 3 | 3 8 4 8 5 | {dsb 6 - - | 5 - - | } 4 - - | 4 - - |
C1: 雪绒花，雪绒花，清晨迎@着@我开放，
C2: E-/ del-/ weiss~, E-/ del-/ weiss~, ev-/ '~r~y morn-/ @ing/ @you/ greet/ me~.
Q2: 3 - 3 | 5 - - | 5 - 5 | 4 - - | 3 - 3 | 3 8 1 8 1 | 1 - - | 2 - - |
Q3: 5&mp -<+ 5 | 5 - -! | 1'> - 1' | 6 - -! | 1' - 1' | 1' 8 6 8 6 | 6 - - | 7 - - |
C3: 雪绒花，雪绒花，清晨迎@着@我开放，
C4: E-/ del-/ weiss~, E-/ del-/ weiss~, ev-/ '~r~y morn-/ @ing/ @you/ greet/ me~.
Q4: 1 - 1 | 7, - - | 3 - 3 | 4 - - | 5 - 5 | 5 8 4 8 3 | 2 - - | 5 - - |
```



Rendered output: Dynamics and raised hairpins in a choir

19. Ornaments, parentheses and annotations

Code after a note	OctoPus sample	Meaning	Current OctoPus rendering
&yc	1	Fermata / sustain	Visible in this OctoPus preview
&bc	1	Tenuto	Visible in this OctoPus preview
&zy	1	Accent	No additional mark in this OctoPus preview

Code after a note	OctoPus sample	Meaning	Current OctoPus rendering
&dy		Staccato	No additional mark in this OctoPus preview
&hx		Breath mark	No additional mark in this OctoPus preview
&shy		Upward slide	Visible in this OctoPus preview
&xhy		Downward slide	Visible in this OctoPus preview
&sby		Upper mordent	Visible in this OctoPus preview
&xby		Lower mordent	No additional mark in this OctoPus preview
&cy		Trill	No additional mark in this OctoPus preview
&sby+		Upper mordent alternate form	Same shape as the base symbol preview
&xby+		Lower mordent alternate form	No additional mark in this OctoPus preview
&cy+		Trill alternate form	No additional mark in this OctoPus preview
&zkh		Opening accompaniment parenthesis	Visible in this OctoPus preview
&ykh		Closing accompaniment parenthesis	Visible in this OctoPus preview

These short codes come from the notation reference, often from Chinese name initials. Attach them to the correct note. The ornament + variants differ from hairpin height; their current shapes and support status are shown in the table. Add a quoted annotation after a note to print text above it. Accompaniment parentheses &zkh and &ykh are also placed after the host digit even when the drawn parenthesis appears before the music.

```
Q: 3&yc 3&bc 3&zy 3&dy | 3&hx 3&shy 3&xhy 3&cy |
Q: 1&zkh 2 3 4&ykh | 5"sing gently" - - - ||
```

Use the correct attachment and variant

&yc marks a fermata, &bc tenuto, &zy accent, &dy staccato, and &hx breath in Jianpu-Spec. &shy/&xhy are slides; &sby/&xby are mordents; &cy is a trill. The table above shows current OctoPus output for each command. Multiple compatible codes can be attached to one host, as in 6/&sby&zkh; check the result for crowding.

Three kinds of text and parentheses

A quoted annotation after a note, such as 5"sing gently", prints near that host; a quoted barline annotation belongs to the boundary; a full # comment never prints. For accompaniment parentheses use &zkh and &ykh after the first and last host. Ordinary (...) in Q makes a musical slur, not a pair of accompaniment glyphs. Ordinary punctuation and the legacy (~...~) form in C belong to lyric text, not music-span syntax.

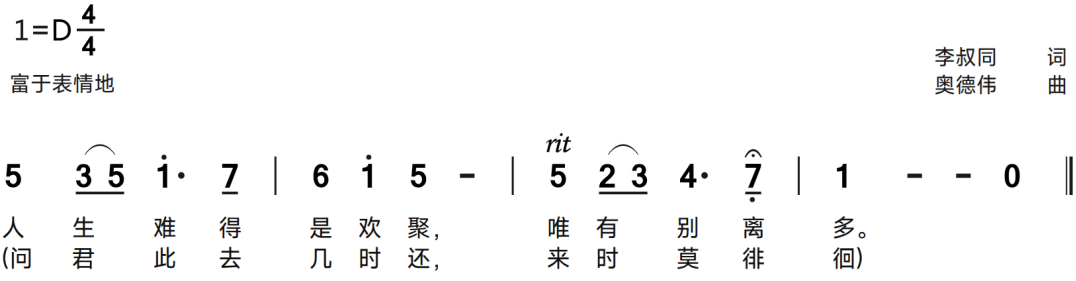
Ritardando and fermata

5&rit prints the tempo instruction. 7,/&yc prints a fermata above the low eighth note. Neither command changes playback, because OctoPus currently has no playback workflow.

bundled examples: Farewell.jps • source lines 34-36. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 5 (3/ 5/) 1'. 7/ | 6 1' 5 - | 5&rit (2/ 3/) 4. 7,/&yc | 1 - - 0 ||
C: 人生@难得是欢聚，唯有@别离多。
C: (~问君@此去几时还，来时@莫徘徊~)
```



Rendered output: Ritardando and fermata

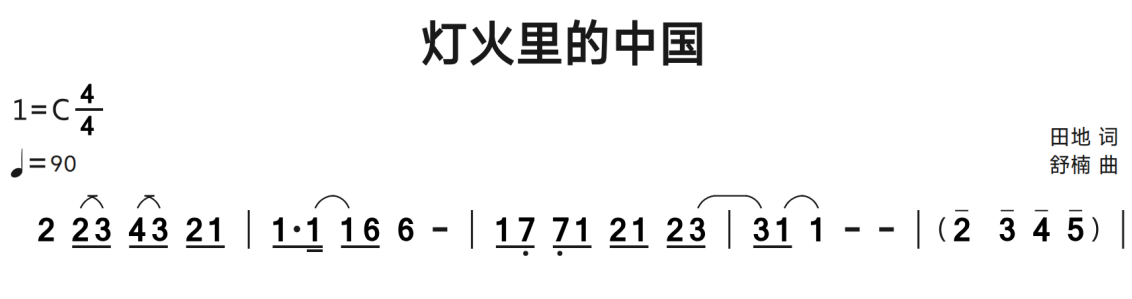
Tenuto and accompaniment parentheses

The final bar attaches &bc to each note. &zkh is attached to its first note and &ykh to its last, creating visible accompaniment parentheses without adding musical events.

bundled examples: China-In-The-Lights.jps • source lines 21-21. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 8 2 (2/ 3/) (4/ 3/) 2/ 1/ | 1/. (1// 1/) 6/ 6 - | 1/ 7,/ 7,/ 1/ 2/ 1/ 2/ (3/ | 3/) (1/ 1) - - | 2&bc&zkh 3&bc 4&bc 5&bc&ykh |
```



Rendered output: Tenuto and accompaniment parentheses

20. Barlines and repeat instructions

Barline code	OctoPus sample	Meaning	Current OctoPus rendering
||/&fine		Fine, end of piece	Text only; no dedicated glyph
||/&dc.		D.C., repeat from the beginning	Text only; no dedicated glyph
:|&ds		D.S., repeat from the segno	Visible in this OctoPus preview
|&ty		Coda sign	Visible in this OctoPus preview
|&hs		Segno sign	Visible in this OctoPus preview

Repeat/navigation marks belong to a barline, not an ordinary note. Use |/ at a row beginning when a barline-attached mark is needed without a visible barline. Use |* to reserve alignment space, especially with mixed single/multiple voices. These marks are printed; playback navigation is not provided.

Visible and invisible boundaries

Use | for an ordinary boundary, || for the final thin/thick boundary and ||/ for two thin lines. |:, :| and :|: mark repeat starts, ends and combined end/start boundaries. A barline divides music; it does not insert a rest. |/ is invisible and takes no barline width, useful at a row-start ending or navigation mark. |* is invisible but reserves width, useful for aligning a later voice entry. These two hidden forms are not interchangeable.

Fine, D.C., D.S., segno and coda

Attach &fine, &dc. and &ds to a barline to print Fine, D.C. or D.S.; &hs draws the segno and &ty the coda sign. For a row-start sign, create a hidden |/ anchor rather than attaching the code to an unrelated note. A repeat ending label such as "1,2" is separate from a navigation command. The viewer displays these symbols but does not execute a playback route. Check that the written route is musically complete yourself.

A repeat beginning inside a source row

The |: occurs after two bars. A repeat boundary is part of the musical flow, not a request for a new source row. The lyric skip markers align unsung notes before the chorus.

bundled examples: JingleBells.jps • source lines 18-20. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 6,/ 4/ 3/~ 8~ 2/ 5/~ 8~ 5/ 5/~ 8~ 5/ | 6/ 5/ 4/~ 8~ 2/ 1. 0/ |: 3/ 3/ 3 3/ 3/ 3 |
C: fun /it /is @/to /ride @/and /sing @/a /sleigh- /ing /song @/to- /night~. @Jin- /gle /bells~, jin- /gle /bells~,
C: 今晚滑@雪真@快乐@把滑雪歌@儿唱。@叮叮当，叮叮当，
```



Rendered output: A repeat beginning inside a source row

Coda and a barline meter annotation

The last |&ty carries a coda sign. The preceding |"p:9/8" introduces a local meter. A quoted barline instruction and an ampersand navigation symbol can serve different purposes on the same kind of anchor.

bundled examples: Autumn-Cicada - Fei-Yuqing.jps • source lines 21-22. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 5 3/ 5 3/ | 1' 6/ 1' 6/ | 5 5/^ (5/ 6/)~ 5/ |"p:9/8" (4/ 5/)~ 4/ (3. 3.) |&ty
C: 秋去冬来美景不在，莫教好@春逝@匆匆，
```



Rendered output: Coda and a barline meter annotation

---

21. Slurs and tuplets

Wrap music in parentheses for a slur: `(1 2 3)`. Spans can nest and continue across barlines or music rows on the same page. Open the span in the earlier Q row and close it in the later row of the same voice. Use `(y1 2 3)` for a tuplet; the group count is inferred from its notes. Do not use a capital Y. Cross-page slurs are not automatically continuous: split/reopen spans at manual page boundaries and inspect the result.

```
Q: (1 (2 3) 4 | 5 6
Q: 7 1') | (y1/ 2/ 3/) 4 5 6 ||
```

Nested, cross-bar and cross-row spans

Open (on the first included note and close) on the last. Nested pairs form separate spans; keep a count while editing. A barline inside the pair does not end the slur. On the same page, leave it open at the end of a Q row and close it on a later row of that voice. If a lyric line intervenes, it remains a lyric, not a slur terminator. At a manual page break, split/reopen the span deliberately. Ending branches also need intentional span boundaries; do not place a close on an unrelated voice merely to silence a warning.

For a continuation that begins before the first note, use a hidden barline inside the parentheses, for example `Q: (( |/ 6)) - - - |`. Each nested pair keeps a separate curve, including a short continuation cap at the row start. A span that crosses music rows draws an open fragment on each row, including when the curved-path style is selected. Every printed inner tie still needs its own parentheses in JPS.

A slur is not a tuplet

(1/ 2/ 3/) is an ordinary three-note slur; (y1/ 2/ 3/) is a tuplet. The renderer infers the count from the group, so changing its notes changes the printed number. Do not type a guessed number after y as though it were a ratio field. Keep the group limited to the intended notes and compare its bracket/number with the reference. A printed tuplet is not proof that every surrounding bar has been rhythmically validated.

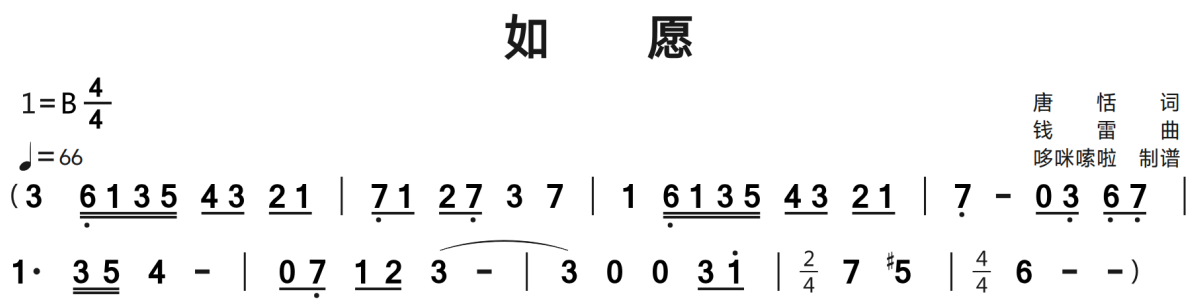
A slur crossing a barline

In the second row, (3 - | 3) opens before a continuation and closes in the next bar. The same note number at both endpoints is not enough: the parentheses define the printed span.

bundled examples: As-Wished.jps • source lines 11-12. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 3&zkh 6,// 1// 3// 5// 4/ 3/ 2/ 1/ | 7,/ 1/ 2/ 7,/ 3 7 | 1 6,// 1// 3// 5// 4/ 3/ 2/ 1/ | 7, - 0/ 3,/ 6,/ 7,/ |
Q: 1. 3// 5// 4 - | 0/ 7,/ 1/ 2/ (3 - | 3) 0 0 3/ 1'/ |"p:2/4" 7 5# |"p:4/4" 6 - -&ykh
```



Rendered output: A slur crossing a barline

Ordinary slurs beside triplet groups

(5// 5) is an ordinary slur; (y1/ 2/ 3/) creates a numbered three-note tuplet. The y is lowercase and comes immediately after the opening parenthesis.

bundled examples: Flower-In-Water.jps • source lines 12-13. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 1'/ 5/ 6/ 3// (5// 5) (y1/ 2/ 3/) | 2. 3/ 6, (y1/ 2/ 3/) | 2. 3/ 6, 0/ 6,/ | 5,/ 3/ 3 - 0 |
C: 凄雨冷风中 @多少繁华如梦 曾经万紫千红 @随风吹落
```



Rendered output: Ordinary slurs beside triplet groups

22. Endings and temporary meter

Ending brackets [and] are attached to barlines. Put their label in a quoted barline annotation. A start at the beginning of a row uses hidden |/. Add / after [for an open right edge; add + (or several +) to raise the bracket and avoid overlapping slurs. Ending spans may cross rows; keep their endpoints in the correct voice. The same brackets attached to a note mean grace notes, so placement matters.

```
Q: |/["1" 1 2 3 4 :|] |/["2" 5 6 7 1' ||]
Q: |/[+/"1,2" 1 2 3 4 | 5 6 7 1' |]
```

For a meter change, put `"p:3/4"` after the barline. This prints the local time signature and updates automatic grouping for the following notes in that voice. Changing to 6/8, 9/8 or 12/8 selects dotted-quarter groups; changing to 3/4 or 4/4 restores quarter-note groups. Use ~ or ^ for any intentional override.

```
Q: 1 2 3 4 |"p:3/4" 5 6 7 |
```

Place each ending on barlines

An ending opens with [attached to a barline and closes with] attached to a later barline. The quoted label belongs to the barline, not to a note. For a beginning at the left of a music row, use |/["1". Use [+, [++ and further + to raise the line above nearby slurs. An open-edge form can use / on the bracket syntax; library scores also use a closing |]/ to omit the right corner. Copy a complete opening/closing pair and inspect it rather than moving brackets alone.

Cross-row endings and local meter

An ending may continue into the next music row on the same page. Keep its label, endpoint, repeat barline and lyric ending consistent. For a local meter, write |"p:3/4" with lowercase p inside the quotes. A row-start local change can use |/"p:3/4". Header P: 3/4 sets the initial meter; if it lists several meters, the first is used until a local change. Within a page, later rows of the same voice retain their local meter. In compound meter, the grouping phase also continues across rows. Other voices retain their own meter, and underline segments end at each row boundary. To return to an earlier meter, write it explicitly, for example |"p:4/4". Reintroduce a local meter after a manual page break when needed. Review durations and custom grouping at every change; a meter annotation does not redistribute the source notes.

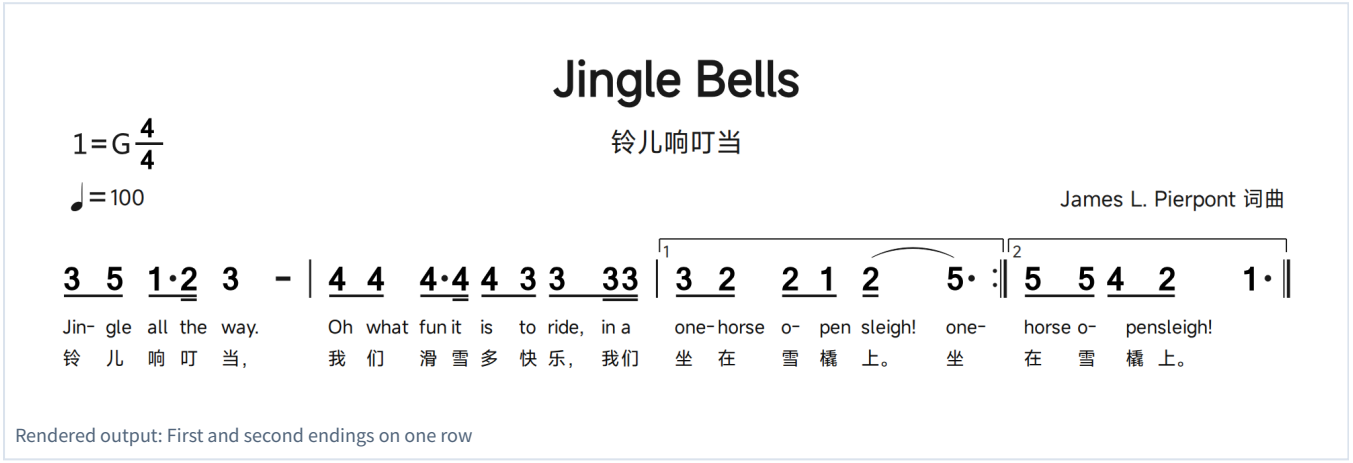
First and second endings on one row

The first ["1" begins on a barline and ends at :|]; ["2" follows on that same boundary and finishes at ||]. Compare the two branches with the matching lyric endings.

bundled examples: JingleBells.jps • source lines 22-24. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 3/ 5/ 1/. 2// 3 - | 4/~ 8~ 4/ 4./~ 4// 4/~ 8~ 3/ 3/~ 3//~ 3// |["1" 3/ 2/ 2/~ 8~ 1/ (2/ 5.) :|]["2" 5/ 5/ 4/~ 8~ 2/ 1. ||]
C: Jin- /gle /all /the /way~. Oh @/what /fun /it /is @/to /ride~, in /a /one- /horse /o- @/pen /sleigh~! one- /horse /o- @/pen
/sleigh~!
C: 铃儿响叮当，我@们滑雪多@快乐，我们坐在雪@橇上。坐在雪@橇上。
```



Rendered output: First and second endings on one row

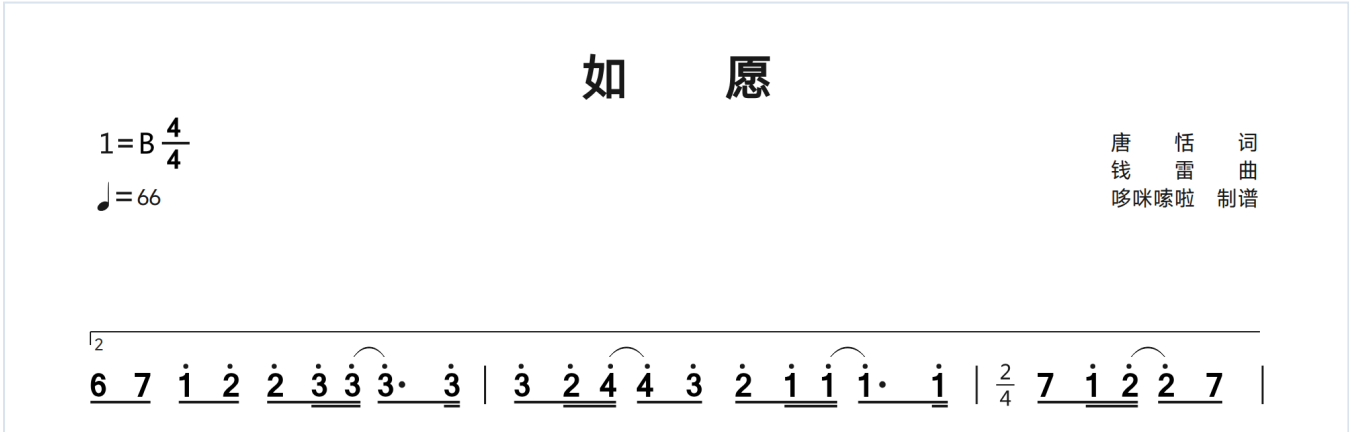
Raised ending, row-start anchor and open right edge

|/ provides an invisible row-start anchor; [+ raises the bracket. The closing |]/ leaves an open right edge. The local "p:2/4" changes the printed meter during the passage.

bundled examples: As-Wished.jps • source lines 32-32. Excerpt rendered with the Linux package font set; blank page space omitted. Extra top clearance separates the isolated ending from the header. Download this JPS excerpt

JPS source

```
Q: |/[+"2" 6/ 7/ 1'/ 2'/ 2'/ 3'// (3'// 3'/.) 3'// | 3'/ 2'// (4'// 4'/) 3'/ 2'/ 1'// (1'// 1'/.) 1'// |"p:2/4" 7/ 1'// (2'// 2'/) 7/ |]/
```



Rendered output: Raised ending, row-start anchor and open right edge

23. Voices and temporary accompaniment

Use Q1, Q2, etc. for numbered music parts; C/C1/C2 rows supply the associated lyric lines. Preserve their ordering and inspect the preview when adapting library scores. A voice name can follow its number in quotes. Keep successive voices rhythmically aligned; numbered music rows form a system, while lyrics attach to their matching voice. Single and multiple voices can be mixed. The spec assigns &sbf to a mid-row voice-brace position; compare its current preview below, where this OctoPus build prints the literal text and leaves the brace at the row start. Use hidden rest 8 and hidden barline |* to reserve earlier space in secondary voices.

```
Q1"Soprano": 1 2 3 4 |
C1: 春 风 吹 来
Q2"Alto": 3, 4, 5, 6, |
C2: 花 儿 开 了
```

Temporary accompaniment uses `{bz ...}`; temporary multiple voices use `{dsb ...}`. The keyword must immediately follow { and be lowercase. These inline groups are positioned against the main music. Copy an appropriate bundled example and check duration/alignment instead of assuming arbitrary blocks will align automatically.

A temporary voice may begin a row, for example `Q: {dsb ( |/ 3 - - - | 3) } ( |/ 1 - - - | 1) 0 0 0 |`. The upper voice follows the written main-voice timeline; an initial hidden barline adds no beats, and a final singleton note remains visible. Following notes return to the main row when the upper voice ends. In a numbered system, the system brace supplies the opening. Keep the main voices' measure durations and separators consistent for shared columns; a missing written barline is not supplied automatically.

A temporary accompaniment block may contain several measures and more notes than its host melody; its notes, visible rests and later measures are retained. In a score with multiple voices, the host melody keeps its shared duration grid with the other parts; the accompaniment does not add beats to that melody. A block at a row end can align with the following row of the same voice. When no following melody is available, including a standalone or terminal block, its contents are still displayed. Nested {bz ...} notes render once. Review alignment whenever the rhythms or measure counts differ.

```
Q: 3 4 | {bz 1&zkh 2 3 4 | 5 6 7 1'&ykh} 5 6 7 1' |
Q: 3 4 | {dsb 1 2 3 4 | 5 6 7 1'} 5 6 7 1' |
```

A minimal mid-row brace pattern (inspect alignment in the preview):

```
Q1: 1 2 3 4 |&sbf 5 6 7 1' |
Q2: 8 8 8 8 |* 3 4 5 6 |
```

Current OctoPus sample of &sbf (the literal text is printed; the brace stays at the row start):

Code	OctoPus sample	Meaning	Current OctoPus rendering
&sbf	sbf | 5 3	Mid-row voice-brace position (spec meaning)	Prints “sbf” as text; brace remains at row start

Build a system one part at a time

Use Q1, Q2, Q3 and Q4 for simultaneous parts, not four successive solo rows. A quoted name follows the number before the colon, for example Q1"Soprano":. Enter one system, compare bar boundaries and durations across its parts, then add its lyric rows and the next system. A missing rest cannot be repaired by visually adding spaces. Library scores include unnumbered C and numbered lyric lanes; their placement is context-sensitive. Preserve a working sample's row ordering and confirm the result when assigning new lyrics.

Temporary blocks and a mid-row brace

{bz ...} is a temporary accompaniment block; {dsb ...} is temporary multiple-voice content. The lowercase keyword follows { immediately. These attach inline to the main passage, while top-level Q2 starts another full part. Count both brace endpoints and the slurs within the block. The spec assigns &sbf to the mid-row brace position. In the current OctoPus preview below, it prints the literal text “sbf” and does not move the brace; treat the code as unsupported for placement in this build. Secondary voices can still use duration-compatible hidden 8 events and |* boundaries before the entry. The minimal pattern is a constructed exercise, not a library quotation.

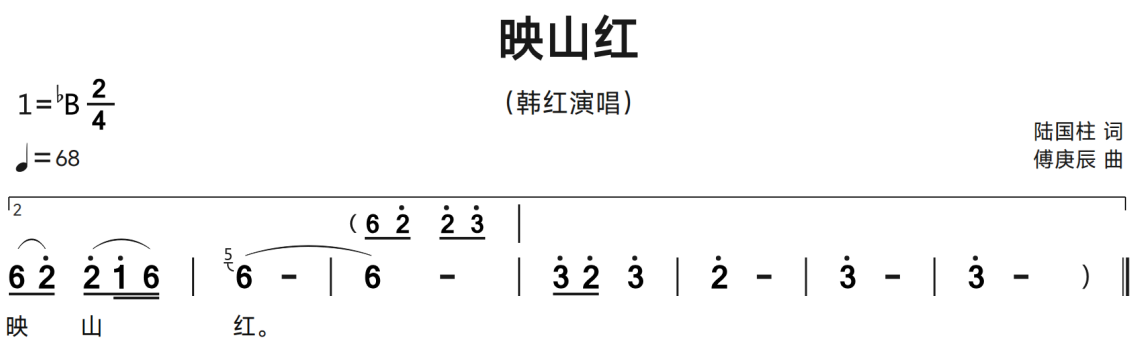
Temporary accompaniment above the melody

The {bz ...} group is embedded inside the second bar, while the main slur closes on the following 6. A raised second-ending bracket spans the passage. Read brace, bracket and parenthesis pairs separately.

bundled examples: Azalea.jps • source lines 29-30. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: |/[+++"2" (6/ 2'/) (2'/ 1'// 6//) | (6[5] - | {bz 6/&zkh 2'/ 2'/ 3'/ | } 6) - | 3'/ 2'/ 3' | 2' - | 3' - | 3' - 8&ykh ||]
C: 映@山@@红。
```



Rendered output: Temporary accompaniment above the melody

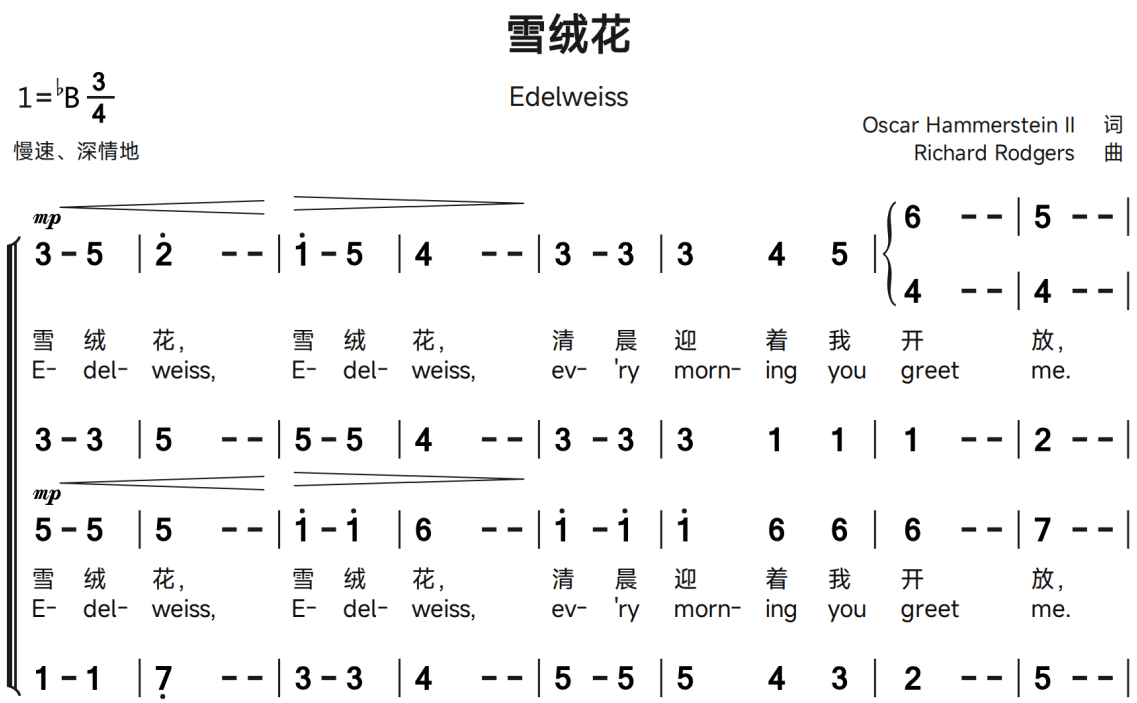
Temporary multiple voices within a full choir system

Q1 contains {dsb 6 - - | 5 - - | } while its main sequence continues with 4. Compare this inline branch with the ordinary Q2/Q3/Q4 parts. A brace group is not a new top-level Q row.

bundled examples: Edelweiss - Choir ChinEng.jps • source lines 11-18. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q1: 3&mp -<+ 5 | 2' - -! | 1'> - 5 | 4 - -! | 3 - 3 | 3 8 4 8 5 | {dsb 6 - - | 5 - - | } 4 - - | 4 - - |
C1: 雪绒花，雪绒花，清晨迎@着@我开放，
C2: E-/ del-/ weiss~, E-/ del-/ weiss~, ev-/ '~r~y morn-/ @ing/ @you/ greet/ me~.
Q2: 3 - 3 | 5 - - | 5 - 5 | 4 - - | 3 - 3 | 3 8 1 8 1 | 1 - - | 2 - - |
Q3: 5&mp -<+ 5 | 5 - -! | 1'> - 1' | 6 - -! | 1' - 1' | 1' 8 6 8 6 | 6 - - | 7 - - |
C3: 雪绒花，雪绒花，清晨迎@着@我开放，
C4: E-/ del-/ weiss~, E-/ del-/ weiss~, ev-/ '~r~y morn-/ @ing/ @you/ greet/ me~.
Q4: 1 - 1 | 7, - - | 3 - 3 | 4 - - | 5 - 5 | 5 8 4 8 3 | 2 - - | 5 - - |
```



Rendered output: Temporary multiple voices within a full choir system

Four aligned parts and reserved blank slots

Q1 through Q4 form one system. Compare the simultaneous barlines and the inserted 8 slots. Numbered C rows in library files can also denote successive lyric lanes: inspect their actual preview rather than mechanically renumbering a borrowed score.

bundled examples: O-Canada Choir.jps • source lines 13-17. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q1: 3 - 5. 5/ | 1 - - 2 | 3 8 4 8 5 6 | 2 - - 0 | 3 - 4#. 4/ | 5 - - 6 |
Q2: 1 - 7,. 7,/ | 6, - - 7, | 1 8 1 8 1 1 | 1 - 7, 0 | 1 - 2. 2/ | 2 - - 3 |
C: O /Can- /a- /da~! Our/ home / @and /@na-tive/ land~! @@True /pa-triot /love/ in
Q3: 5 - 4. 4/ | 3 - - 5 | 5 8 5 8 5 4 | 5 - - 0 | 5 - 1'. 1'/ | 7 - - 6 |
Q4: 1 - 5,. 5,/ | 6, - - 5, | 1 8 2 8 3 4 | 5 - - 0 | 1 - 6,. 6,/ | 7, - - 1 |
```



Rendered output: Four aligned parts and reserved blank slots

24. Lyrics: Chinese, English and mixed text

A C line belongs to the preceding music row; use matching voice numbers for multipart scores. Chinese characters ordinarily map one per note. Punctuation is recognized without becoming a separate sung syllable. @ skips a note; extension - strokes do not each need @. Join two Chinese characters with ~ to fit one note. Several C lines after one Q row supply separate verses.

```
Q: 1 2 3 4 | 5 - 6 7 |
C: 春 风 @ 来 花~儿 开 了
C: 第 二 段 词 也 能 写
```

Begin a lyric line with a quoted label such as "1.". Jianpu-Spec documents _ as a space inside a quoted lyric label, but current OctoPus preserves _ literally and removes ordinary half-width spaces from quoted labels. Use a full-width space if a visible space is needed, and inspect the rendered label. English words/syllables are separated by /; do not map individual letters to notes. / can also make mixed Chinese/English boundaries explicit. Use @ for unsung notes and inspect melismas rather than treating ordinary spaces as dependable English syllable delimiters.

```
Q: 1 2 3 4 |
C: "Verse_1" Sing/a/new/song
C: "2." 春/风/@/song
```

Count sung slots, not visible characters

A Chinese character usually fills one sung slot; punctuation does not require its own note. @ advances past an unsung note, while ~ joins adjacent lyric characters into one slot. A continuation dash does not require another @. When a melody has two notes under one syllable, put the syllable on the first and skip the next as required by the source pattern. Check hidden 8 events and rests as well: they can affect alignment even though a digit is not visible. Add one verse first, then duplicate the C row and replace its words.

Verse labels, spaces and lyric parentheses

Quoted labels precede the lyric they describe, such as C: "1." ... or a label following initial @ skips. In the legacy label convention, use _ for an ordinary space: "Verse_1". Do not confuse these quotes with a note annotation in Q. Some library scores enclose an alternate lyric with (~...~); retain the delimiters when borrowing that pattern and inspect the printed punctuation. Ordinary prose spaces are for readability, not a dependable way of mapping each sung syllable.

English syllables and mixed languages

Use / between sung words or syllables. Split a polysyllabic word deliberately, for example Can-/a-/da, while retaining a visible hyphen where required. A hyphen in C is lyric text, not a musical continuation. @ skips an unsung slot and Chinese/English boundaries may be made explicit with /. The reference allows some separators to be omitted beside Chinese/@, but explicit separators are clearer when editing. Library scores also use ~ within English syllable grouping; inspect the complete pattern before simplifying it.

Chinese verses, skipped notes and a lyric enclosure

Both sung verses follow the same Q row. @ skips the second note of a slurred group. The third C row uses the legacy (~...~) lyric enclosure; this is lyric text/grouping, not a music slur.

bundled examples: Farewell.jps • source lines 17-20. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 5 (3/ 5/) 1'. 7/ | 6 1' 5 - | 5 (2/ 3/) 4. 7,/ | 1 - - - |
C: 晚风@拂柳笛声残，夕阳@山外山。
C: 问君@此去几时还，来时@莫徘徊。
C: (~人生@难得是欢聚，唯有@别离多~)
```



Rendered output: Chinese verses, skipped notes and a lyric enclosure

Chinese joins and bilingual lyric rows

The Chinese text joins 马~儿 and 令~人 so each pair belongs to one lyric slot. The English line uses / to separate words/syllables and @ for unsung notes. Compare both under the same melody.

bundled examples: JingleBells.jps • source lines 14-16. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 5/ 5/ 4/ 2/ (3 1/) 5,/ | 5,/ 3/ 2/ 1/ 5,. 5,/ | 5,/ 3/ 2/ 1/ 6,. 6,/ |
C:  laugh- /ing /all /the /way~.  @@Bells /on /bob- /tail /ring~, @mak- /ing /spir- /its / bright~. What
C: 欢笑又歌唱。@马~儿铃声响叮当，令~人精神多欢畅，我~们
```



Rendered output: Chinese joins and bilingual lyric rows

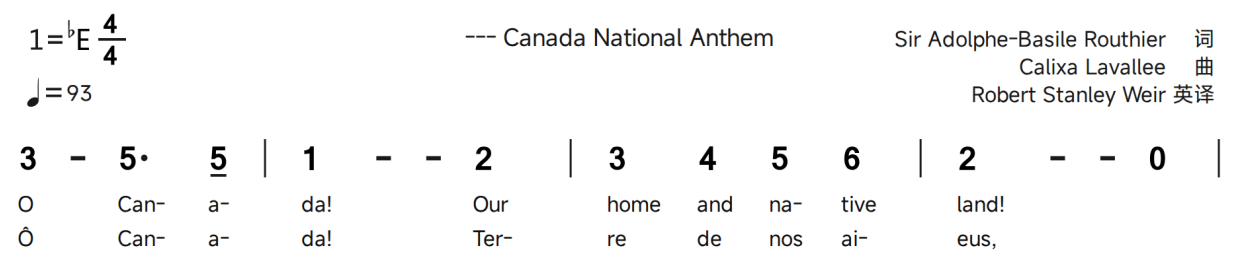
English and French syllable alignment

Can- /a- /da gives three sung slots. A hyphen in a lyric is visible text, not a duration extension. Both C rows share the melody; the long final syllable does not require a new lyric entry for every continuation stroke.

bundled examples: O-Canada.jps • source lines 13-15. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 3 - 5. 5/ | 1 - - 2 | 3 4 5 6 | 2 - - 0 |
C: O /Can- /a- /da~! Our/ home / and /na-tive/ land~!
C: Ô Can- /a- /da~! Ter- /re /de /nos /ai- /eus~,
```



Rendered output: English and French syllable alignment

Quoted verse labels after an initial skip

The pickup rest is skipped with @; then "1.　" and "2.　" label the verses. The library uses a full-width space in those labels. For a portable ASCII-space label, use the documented underscore convention.

bundled examples: As-Wished.jps • source lines 14-16. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q: 0/. 6,// |: 3 1// 2// 3// (4// 4.) 4// 3// | 2/ 1// (2// 2/) 7,/ 3 - | 0 1/ 3/ 2/ 1// (2// 2/.) 1// | 2/ 1// (2// 2/) 7,/ 3. 0// 6,// |
C: @"1.　"你是 遥遥的路　@山野大雾里@的灯　@我是孩童啊 @走在你的@眼眸 @你是 明月清风 @我是
C: @"2.　"@是 岁月长河　@星火燃起的@天空　@我是仰望者 @就把你唱@成歌 @你是
```



Rendered output: Quoted verse labels after an initial skip

25. Manual page breaks

Place the exact lowercase token `[fenye]` on its own line. It creates a new score page; it is not a note annotation. Keep the intended music systems and their lyrics together. Long source does not imply automatic musical reflow or automatic page-spanning slurs. Split/reopen slurs at the page boundary, inspect ending brackets and confirm that each exported page fits its margins.

```
Q: 1 2 3 4 ||
C: 第 一 页 词

[fenye]

Q: 5 6 7 1' ||
C: 第 二 页 词
```

Choose a boundary after a complete system

Put [fenye] on its own line between complete systems. In a multipart score, include every Q part and associated C row before breaking. Close or intentionally split slurs and inspect ending/hairpin endpoints at the boundary. After adding a page, navigate both pages and review their lower margins. The source controls music-row and page boundaries; changing paper or font settings does not replace explicit musical pagination. The example below has two rendered page crops, not a single tall page.

Two systems separated by a manual page break

The first system ends before [fenye]; the next starts with a complete Q1-Q4 system. Both output pages are shown. Keep each system and its lyric rows together when choosing a boundary.

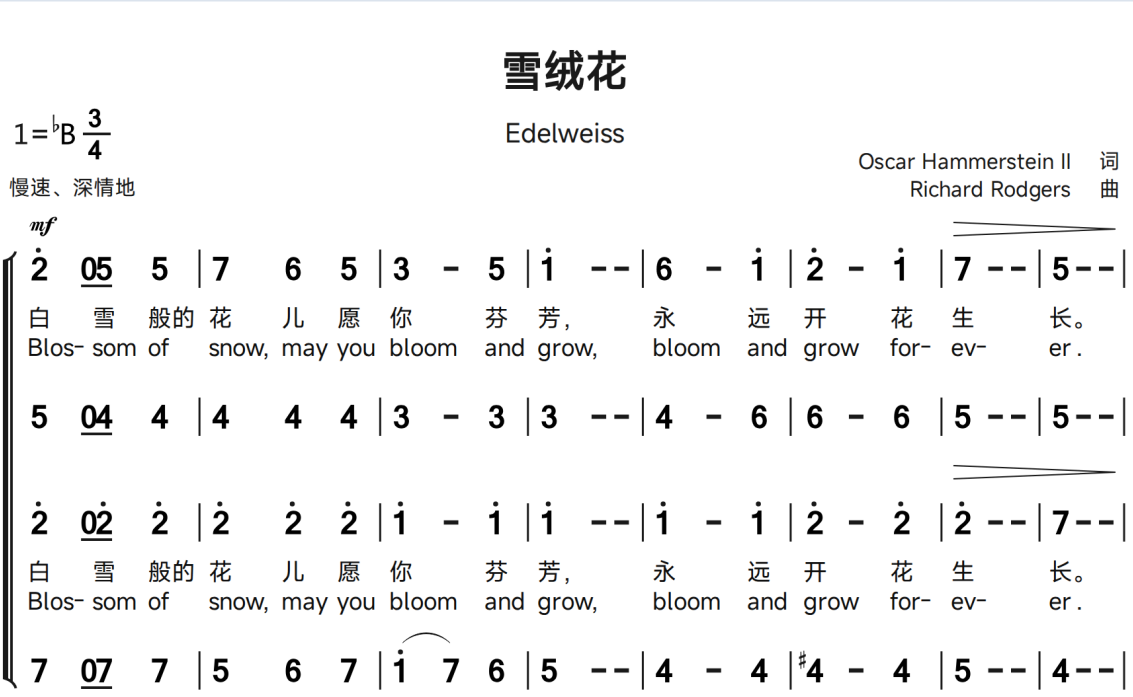
bundled examples: Edelweiss - Choir ChinEng.jps • source lines 29-47. Excerpt rendered with the Linux package font set; blank page space omitted. Download this JPS excerpt

JPS source

```
Q1: 2'&mf 0/ 5/ 8 5 | 7 8 6 8 5 | 3 - 8 5 | 1' - - | 6 - 8 1' | 2' - 8 1' | 7> - - | 5 - -! |
C1: 白@雪@般~的花@儿@愿你@芬芳，永@远开@花生长。
C2: Blos-/ @ som/ @ of/ snow~, @may/ @you/ bloom/ @and/ grow~, bloom/ @and/ grow/ @for-/ ev-/ er.
Q2: 5 0/ 4/ 8 4 | 4 8 4 8 4 | 3 - 8 3 | 3 - - | 4 - 8 6 | 6 - 8 6 | 5 - - | 5 - - |
Q3: 2' 0/ 2'/ 8 2' | 2' 8 2' 8 2' | 1' - 8 1' | 1' - - | 1' - 8 1' | 2' - 8 2' | 2'> - - | 7 - -! |
C3: 白@雪@般~的花@儿@愿你@芬芳，永@远开@花生长。
C4: Blos-/ @ som/ @ of/ snow~, @may/ @you/ bloom/ @and/ grow~, bloom/ @and/ grow/ @for-/ ev-/ er.
Q4: 7 0/ 7/ 8 7 | 5 8 6 8 7 | (1' 7) 8 6 | 5 - - | 4 - 8 4 | 4# - 8 4 | 5 - - | 4 - - |

[fenye]

Q1: 3<+ - 5 | 2' - -! | 1'> - 5 | 4 - -! | 3 - 8 5 | 5 8 6 8 7 | 1'< - - | 1'# - -! |
C1: 雪绒花，雪绒花，永@远祝@福@我家乡。
C2: E-/ del- /weiss~, E-/ del-/ weiss~, bless/ @my/ home-/ @land/ @for-/ ev-/ er~.
Q2: 3 - 3 | 5 - - | 4 - 3 | 4 - - | 3 - 8 3 | 4 8 4 8 4 | 3 - - | 5 - - |
Q3: (3' 2')< 1' | 2' - -! | 1'>++ - 1' | 1' - -! | 1' - 8 1' | 7 8 7 8 2' | 1'<+ - - | 3' - -! |
C3: 雪@绒花，雪绒花，永@远祝@福@我家乡。
C4: E-/ @del- /weiss~, E-/ del-/ weiss~, bless/ @my/ home-/ @land/ @for-/ ev-/ er~.
Q4: 5 - (5/ 6/) | 7$ - - | 6 - 6 | 6$ - - | 5 - 8 5 | 5 8 5 8 5 | 5 - - | 6 - - |
```



Rendered output: Two systems separated by a manual page break (1/2)



Rendered output: Two systems separated by a manual page break (2/2)

26. Practical editing and review checklist

- Choose a bundled sample close to the score: ordinary melody, symbols/grace notes, multiple voices or lyrics. Work on a copy.
- Enter header fields first, then one music row. Check barlines and durations before adding lyrics.
- Attach octave, duration, accidental and ornament suffixes to the intended note. Distinguish note grace brackets from barline ending brackets.
- Add voices and lyric verses one at a time; check alignment after each change.
- Use ~ /^, bracket-height + and page settings deliberately to fix grouping/overlap. They do not repair wrong pitches.
- Compare each transcribed page with its original, including page transitions and annotations.
- Resolve diagnostics, save the editable master, export and inspect the final files.

Use the illustrations as editing exercises

Each illustrated block pairs exact selected source lines with a fresh OctoPus render. The header is included in the download; display crops remove unused page whitespace only. Open the named sample to see its full musical context, or open the excerpt to isolate the syntax. Change one token at a time, compare preview and export, then Undo. A successful render demonstrates notation/layout, not verified transcription or complete harmonic/rhythmic correctness.

27. Troubleshooting and feedback

Symptom	What to check
Unexpected symbols or warnings	ASCII punctuation, colon after the row prefix, balanced quotes/brackets, correct suffix attachment and lowercase keywords.

Symptom	What to check
Lyrics shifted	C/Q voice pairing, @ for skipped notes, ~ for grouped Chinese text, / between English syllables.
Slur/ending overlaps	Correct endpoints, page boundary, manual + offsets and available vertical spacing.
Import fails	Supported file type, one input at a time, 100 MiB/200-page limits, readable unencrypted file and no active conflicting operation.
Transcription unavailable/fails	Packaged OCR worker, original input quality and Poppler availability for PDF processing.
Different font appearance	The selected installed family/substitute, Linux score-font resources, missing characters and external SVG viewer fonts. Open **Help → Font status** to see installation instructions or check again.
Browser or update failure	Default-browser association and network; unavailable release information is not proof of being up to date.

For a reproducible issue, share the app version, OS, minimal JPS, relevant settings, steps and expected/actual result. Include original/reference images only when appropriate. Use Report Issues; proposed changes use the pull-request page. The historical specification’ s old personal contact details are not the OctoPus support route.

28. Reference basis and compatibility

The notation chapters are based on the locally retained Jianpu-Spec v1.0 introduction, header, music, lyrics and feedback sections. Image-only symbol, dynamic and barline tables are presented here as searchable text. This guide adds detailed explanations, 25 attributed Sample Library excerpts and 26 rendered images; original reference screenshots are not redistributed. Short syntax exercises cover cases without a matching library example. Guidance on source formatting and source/preview navigation also draws on the captured Website-Contents/Zhipu-jsCode-editor and was checked against OctoPus. The retained Zhipu-draw file is empty.

The notation syntax comes from the reference specification; product and interface guidance describes OctoPus. Historical references to MIDI, playback, website auto-format settings and personal support contacts do not describe current desktop features. Runtime code, font policy, import limits and sample scores were used to check current behavior. This guide is rewritten for OctoPus rather than copied from the original scripts. The production app includes the guide and its PDFs; references and authoring tools remain in OctoPus-dev.

Jianpu-Spec v1.0 topic map and current support

Specification topic	Manual coverage	Current OctoPus status
Introduction, file structure, header/body order, comments and page breaks (chapters 01–02)	13–14	Q/C structure and page breaks are supported. Blank lines separate systems. V/B/Z/D/P/J headers are covered; X is retained but not rendered.
Music notation: scale degrees, rests, octave, duration, accidentals, dynamics, ornaments, ties/slurs, tuplets, barlines, repeats/endings, grouping and voices (chapter 03)	15–23	Documented notation is supported as described in those sections. Repeat and tempo marks print but do not execute playback. Music ~ and ^ customize beat grouping; they do not alter duration.
Lyrics: C rows, Chinese/English alignment, skips, joins, punctuation and quoted annotations (chapter 04)	24	Supported with the behavior described in section 24. The spec's underscore-as-space convention is not implemented: OctoPus prints `_` literally in quoted labels.
Suggestions and feedback (chapter 05)	27	Feedback guidance is covered. Historical personal contacts and MIDI/audio suggestions are not current product features.

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